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0 Introduction

This report presents a comprehensive dynamic analysis of a single-span metallic truss bridge. The primary objective of this study is to characterize the dynamic behavior of the structure under various loading conditions, ranging from harmonic excitations to seismic events and moving loads.

The analysis is performed using the Finite Element Method (FEM), a discretization technique that allows describing the motion of a continuous system in terms of a finite number of degrees of freedom. The modeling process begins with the definition of an appropriate mesh size, ensuring that the finite elements operate within the quasi-static field relative to the frequency range of interest (0–30 Hz), as required by the specific safety coefficients.

The study is structured into the following key phases:

1. **Modal Analysis:** Extraction of the system's natural frequencies and corresponding modes of vibration by solving the eigenvalue problem for the undamped system, utilizing the structural mass and stiffness matrices derived via the FEM assembly procedure.
2. **Frequency Response Analysis:** Computation of the Frequency Response Functions (FRF) to evaluate the steady-state response of specific points of interest (Nodes A and B) under harmonic forcing.
3. **Seismic Response:** Analysis of the structural response to ground motion, modeled as a time-dependent displacement applied to the bridge constraints (O_1 and O_2), requiring the partition of system matrices to account for constraint motion.
4. **Moving Load Analysis:** Investigation of potential resonance phenomena caused by the transit of a sequence of moving loads, utilizing the modal superposition approach to efficiently handle the dynamic interaction.

Finally, based on the results obtained, a structural modification is proposed to optimize the dynamic performance of the bridge, specifically aiming to reduce vertical acceleration at critical locations while adhering to strict mass constraints.

1 Point 1

The first project phase involved completing all necessary calculations and making the decisions for the bridge's finite element discretization. The details of these calculations, the formulas used, and the resulting decisions are presented hereafter.

1.1 Maximum admissible element length

This is the general way to find the maximum admissible element length considering the safety factor given. The FEM model must be accurate up to a maximum excitation frequency $f_{\max} = 30$ Hz. The corresponding circular frequency is

$$\Omega_{\max} = 2\pi f_{\max} = 2\pi \cdot 30 \text{ rad/s.} \quad (1)$$

The safety factor $s = 1.5$ is applied on this frequency range, so the first bending eigenfrequency ω_1 of each *single* beam element must satisfy

$$\omega_1 \geq s \Omega_{\max} \quad (2)$$

ω_1 depends on the length of the element and on the constraints applied to the beam, for a pin-pin beam:

$$\omega_1(L_k) = \frac{\pi^2}{L_k^2} \sqrt{\frac{E_k J_k}{m_k}}. \quad (3)$$

The element has to behave quasi-statically in the frequency range of interest, this leads to this equation that gives us $L_k \max^1$

$$L_{\max} = \left[\frac{\pi^2}{s \cdot \Omega_{\max}} \sqrt{\frac{E_k \cdot J_k}{m_k}} \right]^{1/2} \quad (4)$$

¹ m_k : mass per unit length $E_k J_k$: bending stiffness

The bridge is all made of steel so the density and the elastic modulus will be the same for all the elements but we have to take into account the different sections.

Steel properties:

$$E = 2.06 \cdot 10^{11} \text{ N/m}^2, \quad \rho = 7800 \text{ kg/m}^3. \quad (5)$$

For each beam family the area A and the second moment of inertia J are given in cm^2 and cm^4 , respectively:

Beam family	Cross section	$A [\text{cm}^2]$	$J [\text{cm}^4]$
Horizontal chords	IPE500	115.5	48200
Diagonal members	HEA300	112.5	18260
Vertical members	IPE240	76.84	7763

Conversion to SI units and mass per unit length:

$$A = A_{\text{cm}^2} \cdot 10^{-4} \text{ m}^2, \quad J = J_{\text{cm}^4} \cdot 10^{-8} \text{ m}^4, \quad m = \rho A. \quad (6)$$

Let's calculate the three different $L_{\max}$.

Top and bottom chords (IPE500).

$$A_{\text{IPE500}} = 115.5 \cdot 10^{-4} \text{ m}^2 = 1.155 \cdot 10^{-2} \text{ m}^2, \quad (7)$$

$$J_{\text{IPE500}} = 48200 \cdot 10^{-8} \text{ m}^4 = 4.82 \cdot 10^{-4} \text{ m}^4, \quad (8)$$

$$m_{\text{IPE500}} = \rho A_{\text{IPE500}} = 7800 \cdot 1.155 \cdot 10^{-2} \text{ kg/m} \simeq 90.1 \text{ kg/m}. \quad (9)$$

$$L_{\max, \text{IPE500}} = \left[\frac{\pi}{2 \cdot 45} \sqrt{\frac{(2.06 \cdot 10^{11})(4.82 \cdot 10^{-4})}{90.1}} \right]^{1/2} \simeq 6.05 \text{ m}. \quad (10)$$

Diagonal beams (HEA300).

$$A_{\text{HEA300}} = 112.5 \cdot 10^{-4} \text{ m}^2 = 1.125 \cdot 10^{-2} \text{ m}^2, \quad (11)$$

$$J_{\text{HEA300}} = 18260 \cdot 10^{-8} \text{ m}^4 = 1.826 \cdot 10^{-4} \text{ m}^4, \quad (12)$$

$$m_{\text{HEA300}} = \rho A_{\text{HEA300}} = 7800 \cdot 1.125 \cdot 10^{-2} \text{ kg/m} \simeq 87.8 \text{ kg/m}. \quad (13)$$

$$L_{\max, \text{HEA300}} = \left[\frac{\pi}{2 \cdot 45} \sqrt{\frac{(2.06 \cdot 10^{11})(1.826 \cdot 10^{-4})}{87.8}} \right]^{1/2} \simeq 4.78 \text{ m}. \quad (14)$$

Vertical beams (IPE240).

$$A_{\text{IPE240}} = 76.84 \cdot 10^{-4} \text{ m}^2 = 7.684 \cdot 10^{-3} \text{ m}^2, \quad (15)$$

$$J_{\text{IPE240}} = 7763 \cdot 10^{-8} \text{ m}^4 = 7.763 \cdot 10^{-5} \text{ m}^4, \quad (16)$$

$$m_{\text{IPE240}} = \rho A_{\text{IPE240}} = 7800 \cdot 7.684 \cdot 10^{-3} \text{ kg/m} \approx 59.9 \text{ kg/m}. \quad (17)$$

$$L_{\max, \text{IPE240}} = \left[\frac{\pi}{2 \cdot 45} \sqrt{\frac{(2.06 \cdot 10^{11})(7.763 \cdot 10^{-5})}{59.9}} \right]^{1/2} \simeq 4.25 \text{ m}. \quad (18)$$

1.2 Finite element mesh design

In the geometry of the bridge given, the distance between two consecutive joints along the horizontal chords is:

$$L_{\text{horiz}} = 5 \text{ m},$$

while the vertical members have a height

$$L_{\text{vert}} = 4 \text{ m}.$$

Both values are below the corresponding maximum admissible lengths so it is sufficient to use one element per horizontal and vertical member between two joints.

Each diagonal member connects two joints separated horizontally by about 5 m and vertically by about 4 m, leading to an actual diagonal length is:

$$L_{\text{diag}} \approx \sqrt{5^2 + 4^2} \text{ m} \simeq 6.4 \text{ m} > L_{\text{max,HEA300}} \approx 4.78 \text{ m}.$$

This exceeds the maximum admissible element length for HEA300 sections, to satisfy the frequency requirement, each diagonal is therefore split into two elements by inserting a node in the middle.

The bridge has been discretised with the following criteria:

- a node is placed at each *real joint* where members meet (supports, intersections between chords, verticals, and diagonals);
- horizontal chords and vertical members are modelled with a single beam element between two consecutive joints;
- diagonal members are subdivided by introducing an additional *mid-node* so that each diagonal is represented by two elements in series.

In summary:

- horizontal chords (IPE500): $L = 5 \text{ m} < L_{\text{max,IPE500}} \simeq 6.05 \text{ m}$;
- vertical members (IPE240): $L = 4 \text{ m} < L_{\text{max,IPE240}} \simeq 4.25 \text{ m}$;
- diagonal members (HEA300): $L = 3.2 \text{ m} < L_{\text{max,HEA300}} \simeq 4.78 \text{ m}$.

The final node layout and numbering adopted in the FEM model are shown in Figure 1.

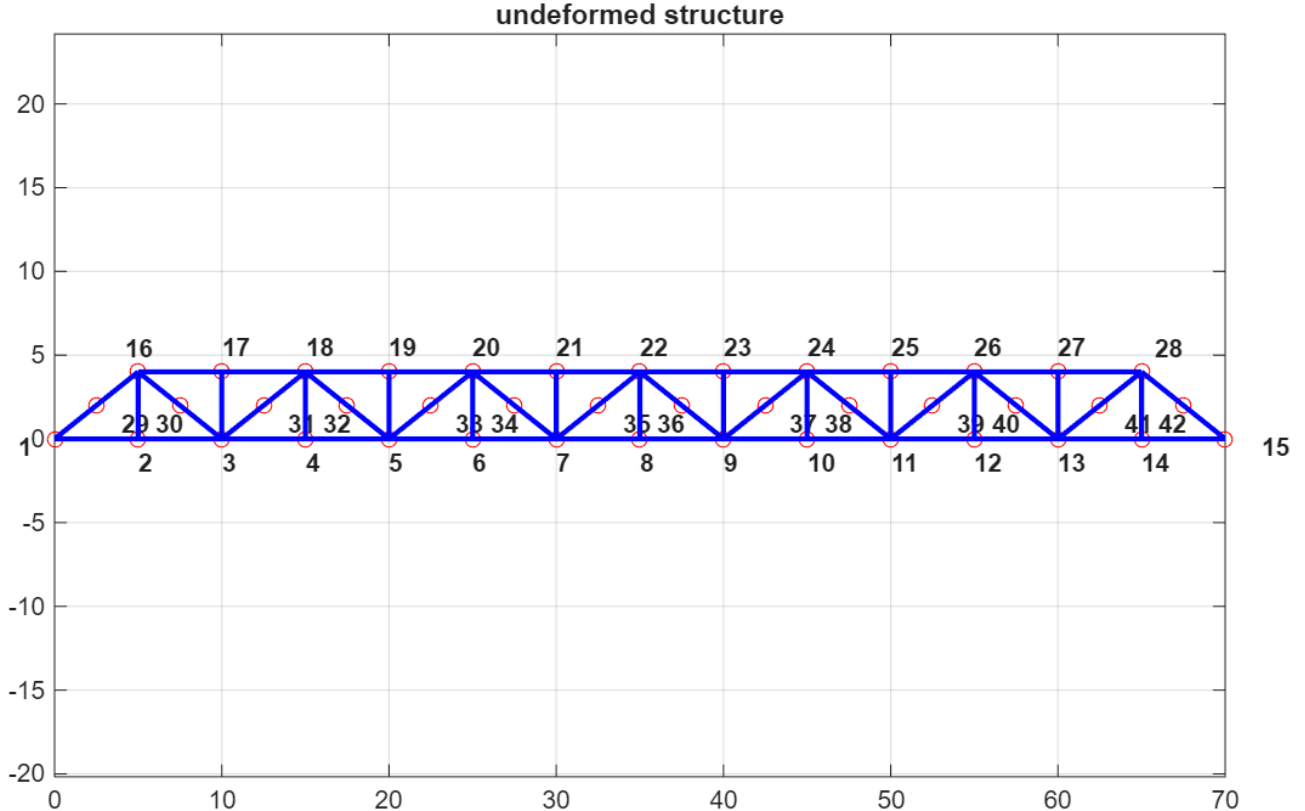


Figure 1: Undeformed configuration of the bridge with node positions and numbering.

1.3 Bending stiffnesses

For each beam family the following quantities are also needed in the FEM input:

- Axial stiffness EA [N],
- Bending stiffness EJ [N m²].

Top and bottom chords (IPE500)

- Axial stiffness:

$$EA_{\text{IPE500}} = 2.06 \cdot 10^{11} \cdot 0.01155 \approx 2.38 \cdot 10^9 \text{ N.} \quad (19)$$

- Bending stiffness:

$$EJ_{\text{IPE500}} = 2.06 \cdot 10^{11} \cdot 4.82 \cdot 10^{-4} \simeq 9.93 \cdot 10^7 \text{ N m}^2. \quad (20)$$

Diagonal beams (HEA300)

- Axial stiffness:

$$EA_{\text{HEA300}} = 2.06 \cdot 10^{11} \cdot 0.01125 \simeq 2.32 \cdot 10^9 \text{ N.} \quad (21)$$

- Bending stiffness:

$$EJ_{\text{HEA300}} = 2.06 \cdot 10^{11} \cdot 1.826 \cdot 10^{-4} \approx 3.76 \cdot 10^7 \text{ N m}^2. \quad (22)$$

Vertical beams (IPE240)

- Axial stiffness:

$$EA_{\text{IPE240}} = 2.06 \cdot 10^{11} \cdot 0.007684 \approx 1.58 \cdot 10^9 \text{ N.} \quad (23)$$

- Bending stiffness:

$$EJ_{\text{IPE240}} = 2.06 \cdot 10^{11} \cdot 7.763 \cdot 10^{-5} \approx 1.60 \cdot 10^7 \text{ N m}^2. \quad (24)$$

Summary The final element properties used in the FEM model are:

Beam family	m [kg/m]	EA [N]	EJ [N m ²]
Horizontal chords (IPE500)	90.1	$2.38 \cdot 10^9$	$9.93 \cdot 10^7$
Diagonal members (HEA300)	87.8	$2.32 \cdot 10^9$	$3.76 \cdot 10^7$
Vertical members (IPE240)	59.9	$1.58 \cdot 10^9$	$1.60 \cdot 10^7$

2 Point 2

2.1 Frequencies

Natural frequencies represent how our system would vibrate if it were left free to oscillate. In a general case, in order to evaluate them, we need to solve the following expression (for n-dof systems):

$$[M]\ddot{x} + [K]x = 0 \quad (25)$$

where $[M]$ and $[K]$ are respectively the mass and stiffness matrices of the system. We can highlight how it does not take into consideration neither external forces nor the damping matrix. In addition we can state:

$$x = x_0 e^{i\omega t} \quad (26)$$

$$\dot{x} = i\omega x_0 e^{i\omega t} \quad (27)$$

$$\ddot{x} = -\omega^2 x_0 e^{i\omega t} \quad (28)$$

These equations lead to;

$$(-\omega^2[M] + [K])x_0 = 0 \quad (29)$$

Our natural frequencies are all the values of ω that verify this result:

$$\det(-\omega^2[M] + [K]) = 0 \quad (30)$$

We are excluding the case where $x_0 = 0$, since it is trivial and useless for our analysis. However, we need to extend these considerations to a generic case, where we have constraints in the system (in our bridge, the hinge in node 1 and the cart at the end in node 15). The resolution is the same, but we need to consider just the $[M_{ff}]$ and $[K_{ff}]$ matrices. They are constituted by the d.o.f that are free to move and are not limited by a constraint: we will exclude the x and y movement in the node 1 and the y movement in the node 15. The resolution is still the same:

$$\det(-\omega^2[M_{ff}] + [K_{ff}]) = 0 \quad (31)$$

In the frequency range 1:30 Hz, only the first 5 natural frequencies are present:

Table 1: First Natural Frequencies (Hz)

Mode (n)	Frequency (ω_n) [Hz]
1	2.31987
2	8.31200
3	12.72639
4	17.65917
5	26.46895

2.2 Modes of vibration

To each natural frequency, it is associated a relative mode of vibration, which shows how the system will move at that frequency. It corresponds to the value of x_0 which satisfies the following equation:

$$(-w_i^2[M] + [K])x_{0i} = 0 \quad (32)$$

Where w_i is the i -th natural frequency of the system. The modes of vibrations are vectors 123×1 , and we are going to consider the ones relatives to the first 5 modes. They can be shown in the following figures (where a scale factor of 10 is being considered to have useful visual results):

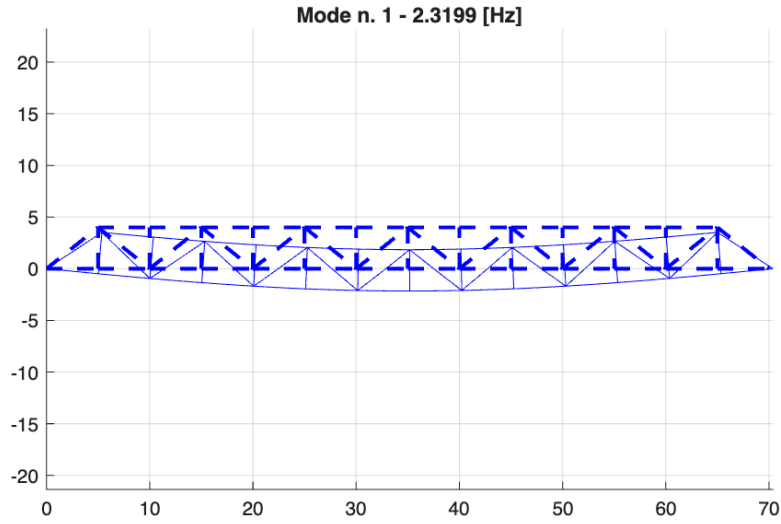


Figure 2: First mode of vibration

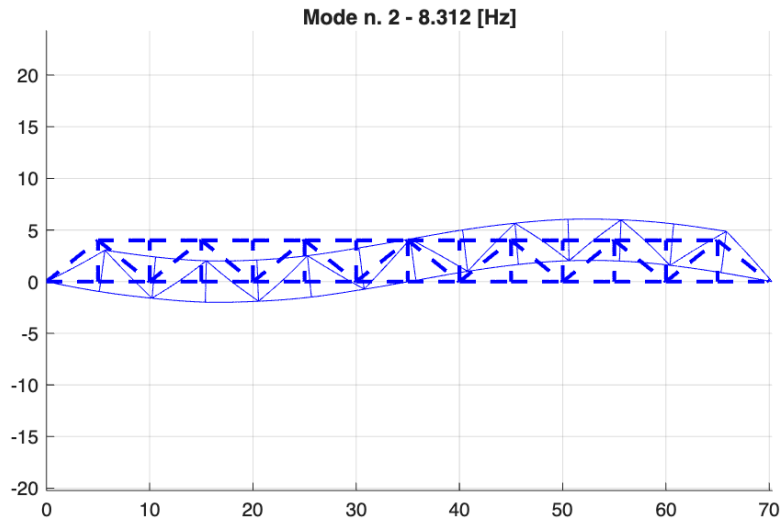


Figure 3: Second mode of vibration

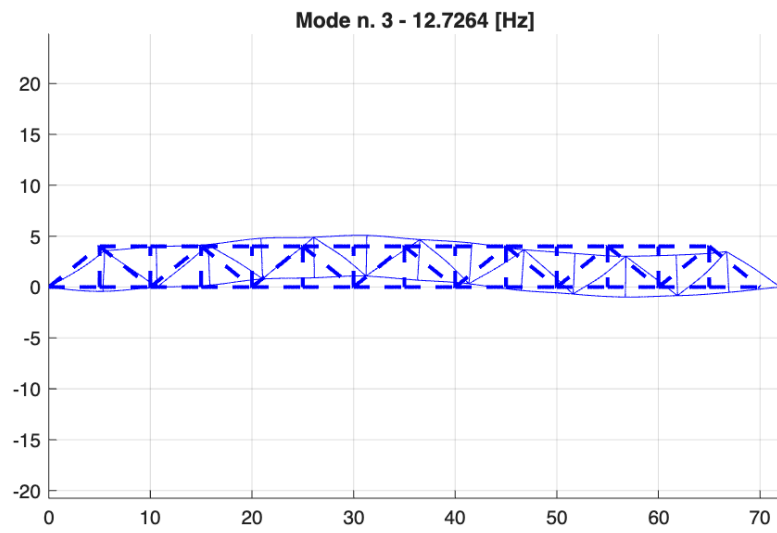


Figure 4: Third mode of vibration

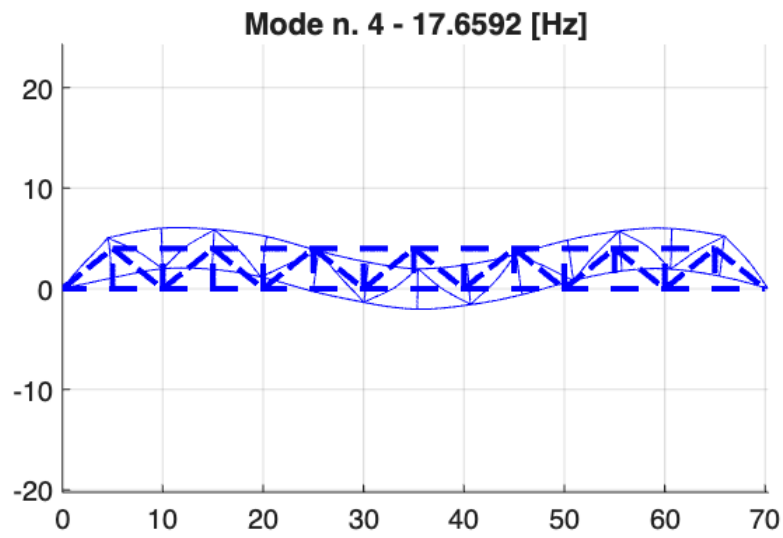


Figure 5: Fourth mode of vibration

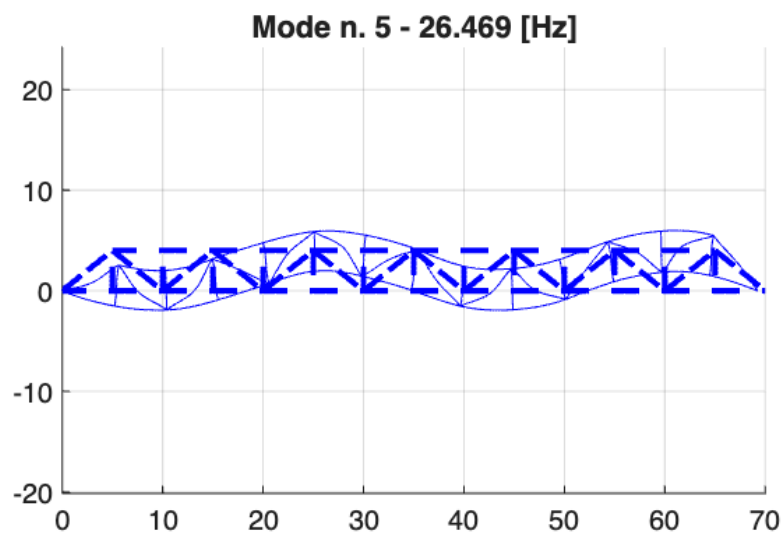


Figure 6: Fifth mode of vibration

2.3 Comments and analysis of results

- Numerical tests performed on different computers yielded consistent mode shapes, though some results were inverted (mirrored on the x-axis). This inversion is a standard property of eigenvalue problems: the eigenvectors representing the mode shapes are scale-invariant. Therefore, a shape and its inverse are physically equivalent solutions, representing the same oscillation limits.
- We plotted only the first 5 modes because their natural frequencies fall below the maximum frequency limit $\Omega_{\max}$ of 30 Hz established for our analysis. Consequently, the contribution of modes with eigenfrequencies higher than $\Omega_{\max}$ has been deemed negligible for the purpose of this evaluation.
- Definition of a Nodal Point(Node): A point on a vibrating body that remains stationary while the rest of the body is undergoing free vibration at a specific natural frequency. The displacement at a node is always zero.

Modal Complexity and Nodal Points: In the free vibration analysis of the structural system performed using the Finite Element Method, a direct correlation is observed between the mode number (n) and the number of intermediate nodal points present in the corresponding mode shape. The n^{th} mode of vibration is characterized by the presence of $(n - 1)$ intermediate nodes.

- Implication for Frequency Response Function (FRF) Analysis: This observation is critical for the next phase of our work (FRF calculation). In fact the location where we apply the force determines which modes are activated. If you apply an excitation force exactly at a nodal point of a specific mode, you cannot excite that mode. The structure will vibrate, but that specific mode shape will effectively be "invisible" in the response. The physical reason is that the mechanical energy transfer requires displacement (Work=Force×Displacement). Since a nodal point has zero displacement for that specific mode, the force performs zero work on it. No energy is transferred into that mode.
- Symmetry properties of the mode shapes. **Symmetry considerations and limitations.** The bridge is *geometrically* symmetric with respect to the mid-span vertical axis and it is supported through vertically symmetric boundary conditions (hinge at O_1 , roller at O_2). As a consequence, the *lowest* vibration modes tend to resemble the classical symmetric/antisymmetric patterns of an equivalent simply-supported beam.

However, since the structure is a **discrete truss/beam assemblage** (different member families, local stiffness variations, and finite-element discretization), this classification must be interpreted as **approximate** and primarily valid for the **first few global modes**. As the mode order increases, **local deformation mechanisms** and member-level contributions become more relevant, and the mode shapes may deviate from "pure" symmetry/antisymmetry.

Predominantly symmetric modes (typically the 1st, 3rd, 5th global bending-like modes). In these modes, the deformation pattern is approximately mirrored about the mid-span axis. Point A (mid-span) typically behaves as an **anti-node** (large vertical displacement), and the rotation/slope at mid-span is usually small (nearly zero), consistent with a symmetric global bending shape.

Predominantly antisymmetric modes (typically the 2nd, 4th, global bending-like modes). In these modes, the deformation on the two halves of the bridge is approximately opposite in sign. As a result, Point A tends to be **close to a nodal location** (small vertical displacement). Nevertheless, due to the discrete nature of the truss and the presence of local deformation components, the mid-span displacement is not necessarily exactly zero for higher-order modes; therefore, the "node at mid-span" interpretation should be regarded as an approximation that holds well mainly for the lower antisymmetric global modes.

- Fundamental Frequency (f_1): After research, we found that the value of 2.32 Hz is typical for bridges with a span of this magnitude (approximately 70 m). This relatively low frequency indicates a flexible structure. This characteristic is particularly critical regarding dynamic response, as it falls within the range of high spectral content for seismic events (typically 0.5 – 5 Hz) and is also relevant for the dynamic interaction with moving train loads.

3 Point 3

3.1 Theoretical Overview

The Frequency Response Function (FRF) describes the steady-state response of a linear structure to a harmonic excitation. While the FRF can theoretically be obtained by directly inverting the matrix $([K] - \omega^2[M] + i\omega[C])$, this is computationally expensive for complex models.

Therefore, the Modal Superposition Method is commonly employed. This technique decouples the system of equations using the structure's natural modes (Φ) and frequencies (ω_n). The total response is calculated by summing the individual contributions of each mode, weighted by the participation of that mode in the applied force and the output location. This approach is highly efficient and provides physical insight into which modes dominate the response at specific frequencies.

3.2 Computational Procedure

The analysis was performed using the MATLAB software `dmb_fem2`, following these steps:

1. **Model Assembly:** The global mass $[M]$ and stiffness $[K]$ matrices were assembled using the provided geometric and material inputs.
2. **Damping Definition:** Structural damping was modeled according to the Rayleigh proportional damping assumption ($[C] = \alpha[M] + \beta[K]$), using coefficients $\alpha = 0.15$ and $\beta = 9.0e-5$.
3. **FRF Computation:** The software computed the response in the $0 \div 30$ Hz frequency range with a step of 0.01 Hz. The code utilizes the eigenvectors computed in the modal analysis phase to synthesize the response functions (Modal Superposition), obtaining: Vertical displacement per unit force (m/N) and vertical acceleration per unit force ($m/s^2/N$).

3.3 Comments and analysis of results

Remarks: The analysis focuses on points A and B, which correspond to node 8 and node 5 of the model, respectively.

The first figures were obtained using the linear plot of the FRFs because we have established that the linear plot is excellent for precisely determining the frequency and the maximum amplitude of the dominant resonance. In contrast, the logarithmic plot is useful for identifying all vibration modes and for understanding the global behavior of the system across the entire frequency spectrum. In fact, while the linear plot is limited to a narrow interval, the logarithmic plot clearly shows:

- The very low frequency behavior (below 1 Hz), which is often correlated with the static stiffness or the rigid body behavior of the system.
- The high frequency behavior (above 30 Hz), which is typically controlled by the mass (or inertia).

We have, however, included the Frequency Response Function (FRF) plots on a logarithmic scale (each figure corresponding to the linear-scale FRF plot). This was done to clearly demonstrate the visibility of all peaks and to emphasize the suitability of the logarithmic scale for understanding the overall system behavior.

3.3.1 FRF 1 (see figure 7) and FRF 2 (see figure 9):

The spectral shape of a Frequency Response Function—specifically the prominence of the fundamental mode and the attenuation or absence of higher modes—is governed by two universal mechanical principles: Modal Controllability and Dynamic Stiffness Scaling.

1. **Modal Controllability:** As we said before, for a vibration mode to be excited, the external force must perform work on the structure. If the excitation is applied at a nodal point, the modal participation factor becomes zero. Consequently, even if the system possesses a natural frequency at that value, the mode remains unexcited and "invisible" in the frequency response.
2. **Dynamic Stiffness Scaling:** Even when modes are successfully excited, the displacement amplitude typically exhibits a decaying trend across the frequency spectrum, in fact the higher-order modes correspond to higher energy deformation states, characterized by greater curvature and significantly higher dynamic stiffness.

The difference in amplitude between the plots is dictated by the mode shapes. For the first mode (symmetric), Point A is close to the mid-span anti-node (maximum displacement), resulting in a higher amplitude compared to Point B, which is located at approximately 1/4 of the span. Conversely, for the second mode (antisymmetric), Point B is closely collocated to an anti-node (bigger displacement), whereas Point A is close to a nodal location for the (predominantly) antisymmetric global mode(s), hence the modal participation is strongly reduced, thus exhibiting significantly minor displacement (not visible in the mode diagrams).

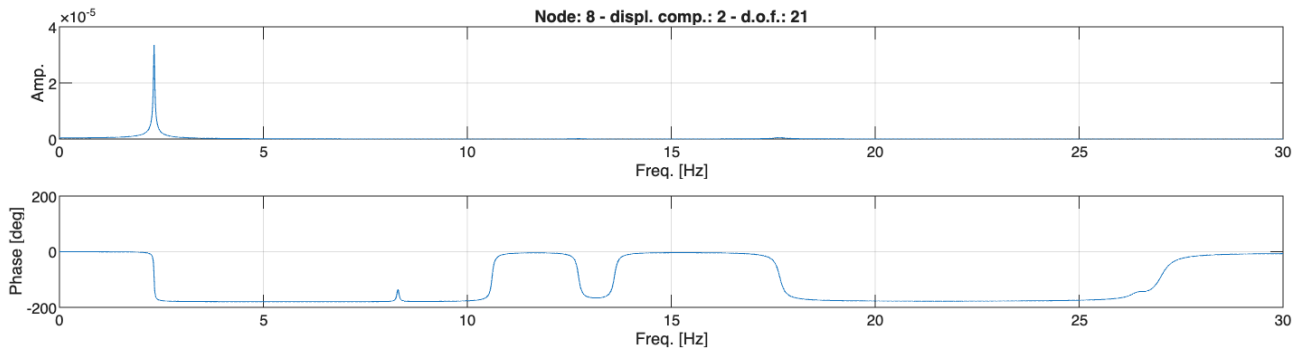


Figure 7: Vertical displacement of point A produced by a vertical force on point A.

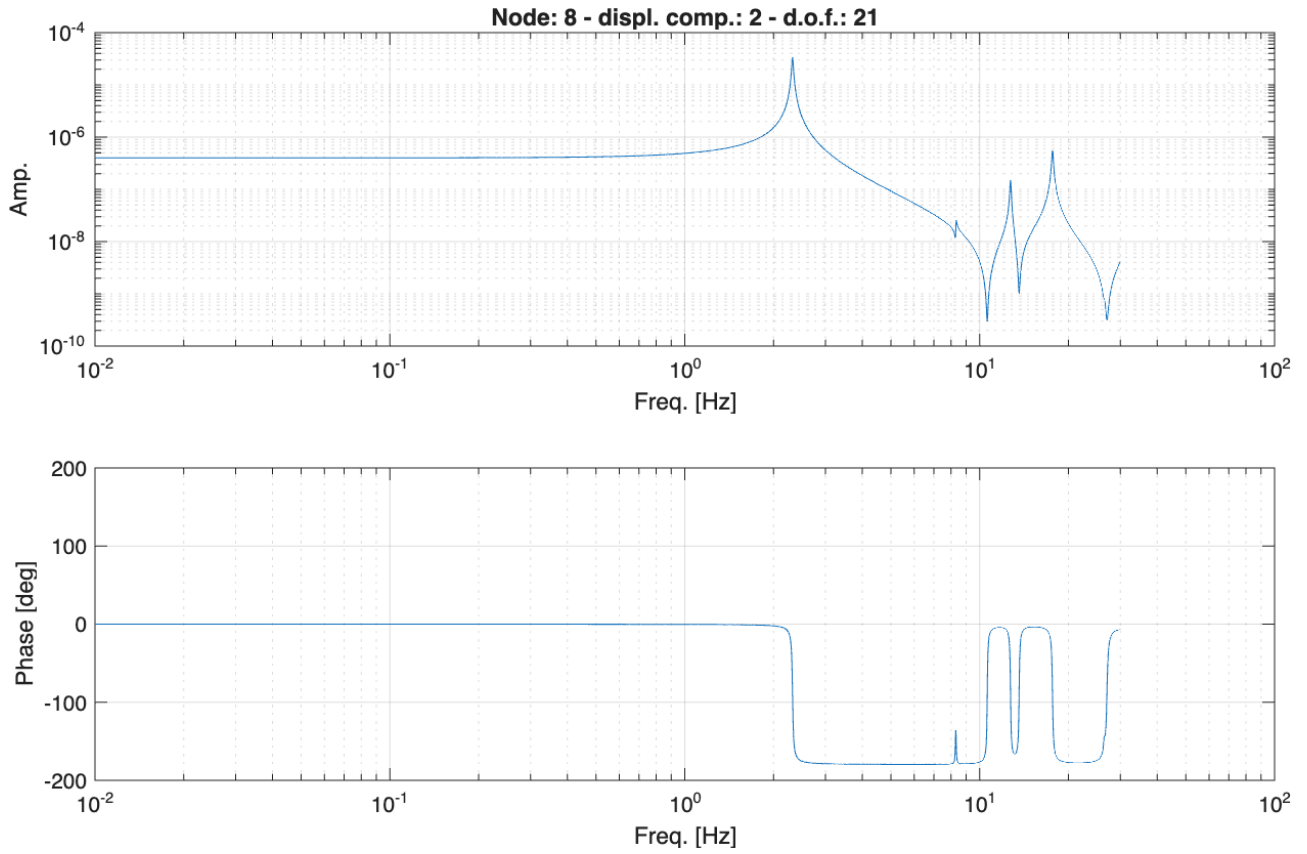


Figure 8: Vertical displacement of point A produced by a vertical force on point A - Log scale

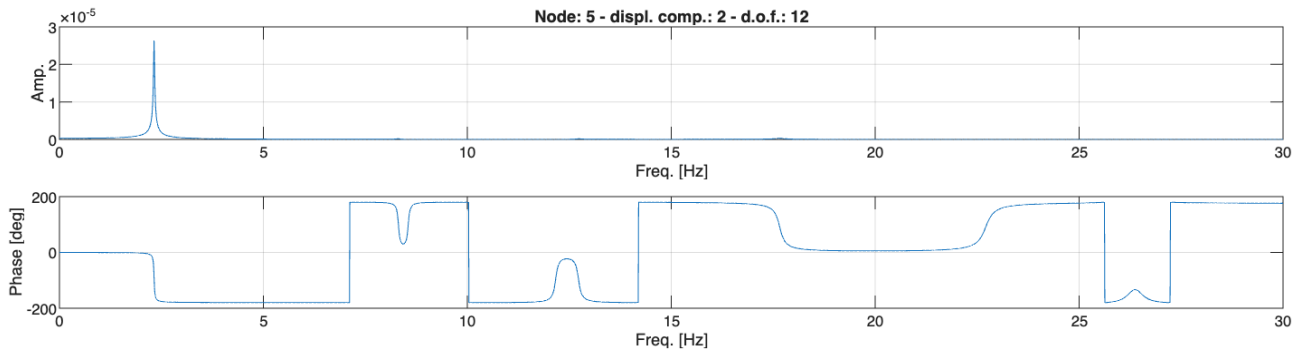


Figure 9: Vertical displacement of point B produced by a vertical force on point A.

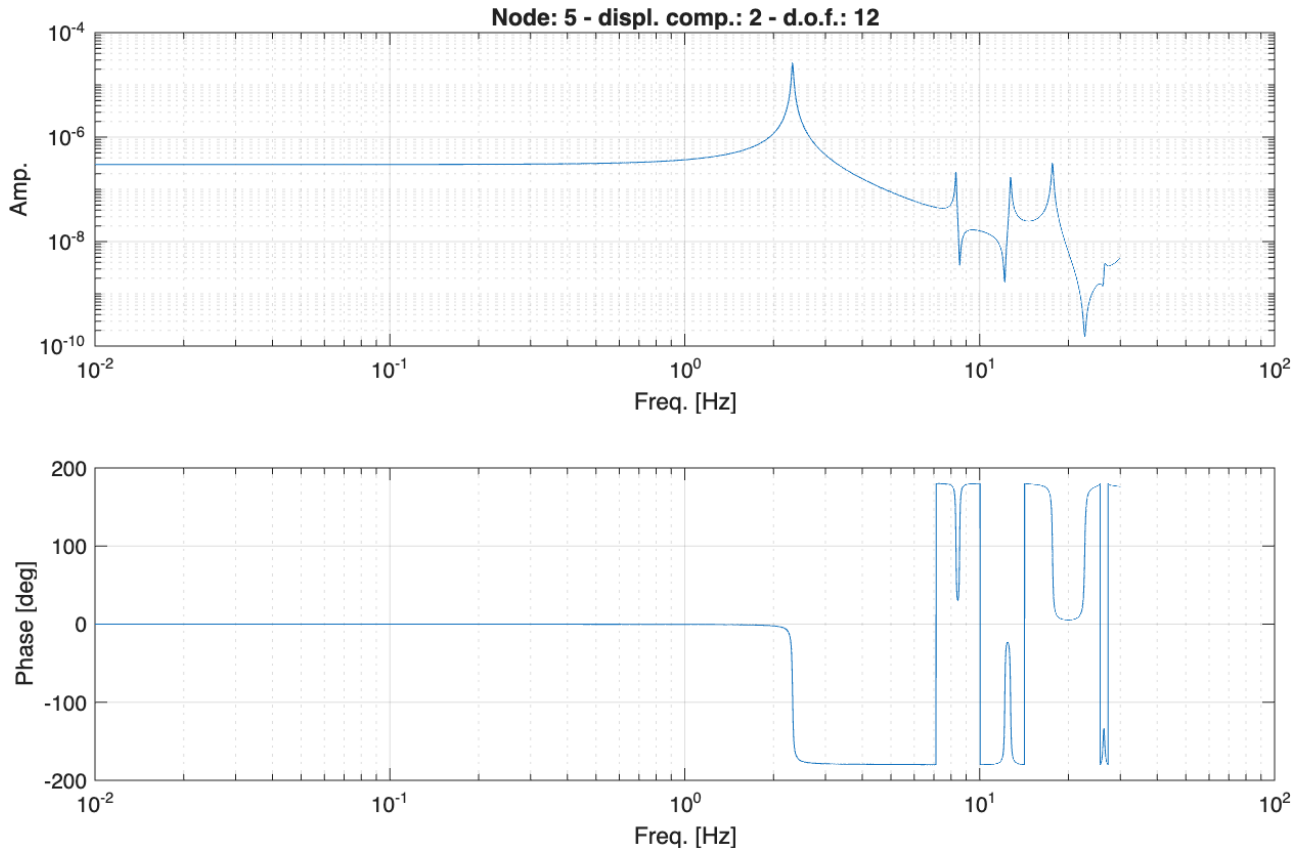


Figure 10: Vertical displacement of point B produced by a vertical force on point A - Log scale

3.3.2 FRF 3 (see figure 12) and FRF 4 (see figure 14):

Analysis of Acceleration FRF with force applied at Point B

The final shape of the FRF is a combination of the frequency effect (ω^2) and the spatial effect (Mode Shape).

Unlike the mid-span excitation, applying force at Point B (an off-center location) excites all modes. Point B is not a nodal point for the antisymmetric modes; therefore, peaks corresponding to Mode 2 (8.3 Hz) and Mode 4 (17.6 Hz) are clearly visible in the response spectrum. The magnitude of a resonance peak depends heavily on where you apply the force and measure the response relative to the "peaks and valleys" (anti-nodes and nodes) of that specific mode shape.

The relative magnitude of the peaks differs significantly from displacement data. While displacement is dominated by the fundamental mode, the acceleration response ($A = -\omega^2 * X$) inherently emphasizes higher frequencies. This explains why the peak for Mode 2 appears significantly "more relevant" (higher amplitude) than Mode 1 in the acceleration FRF, despite Mode 1 having larger physical displacement.

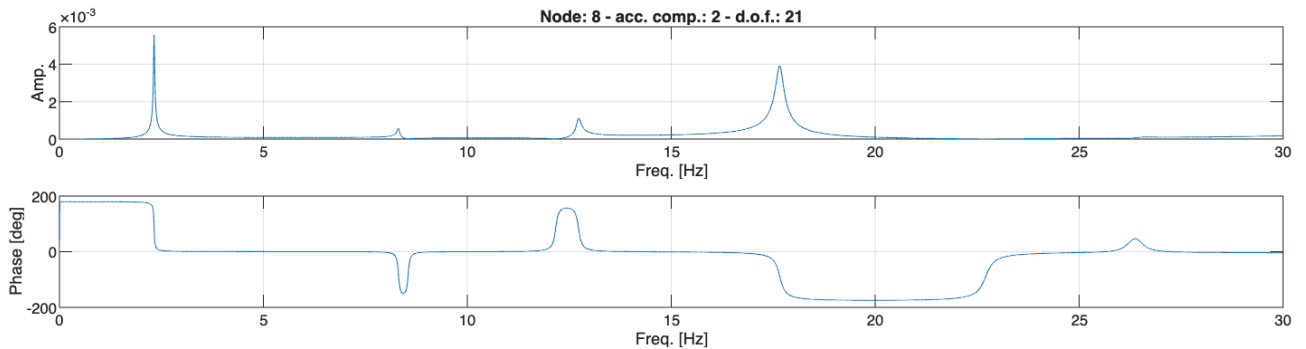


Figure 11: Vertical acceleration of point A produced by a vertical force on point B.

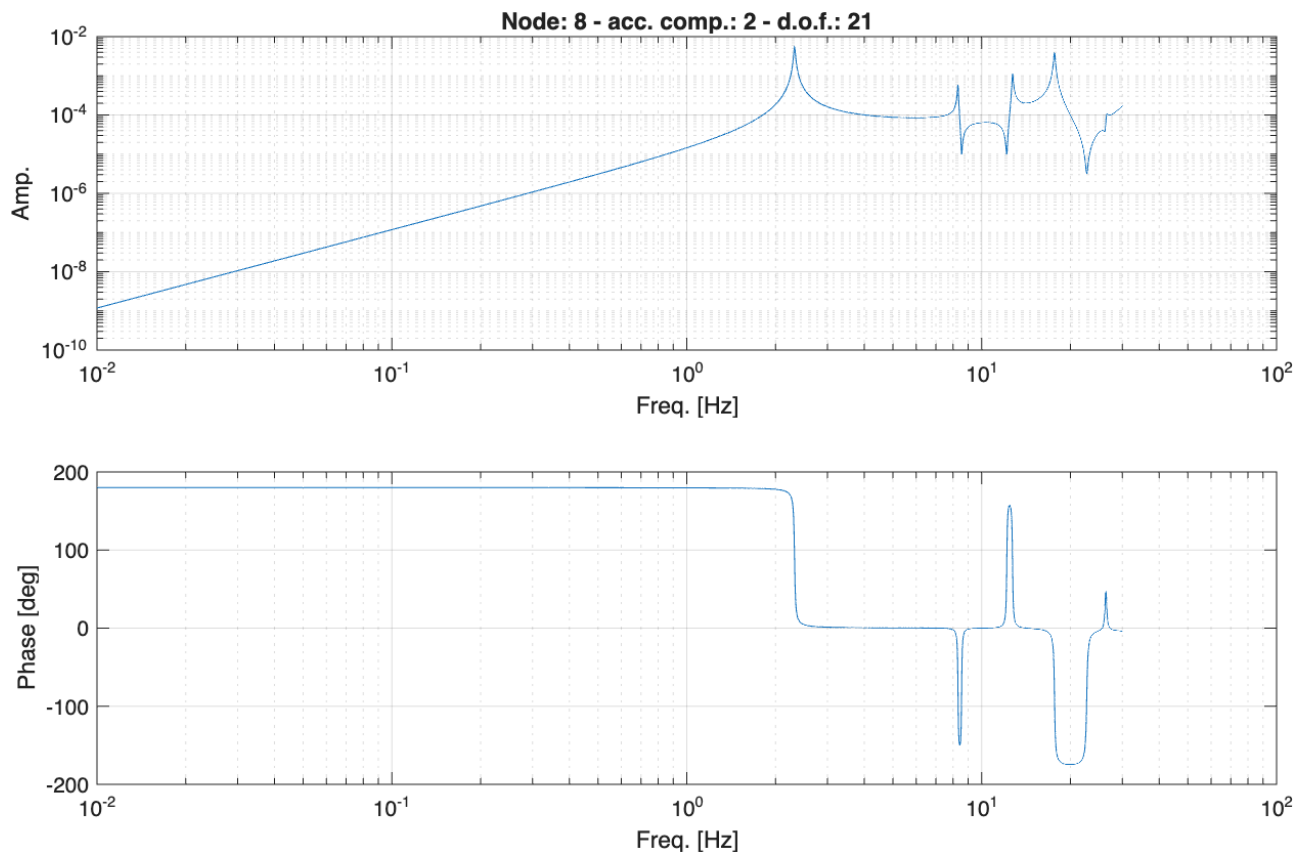


Figure 12: Vertical acceleration of point A produced by a vertical force on point B - Log scale

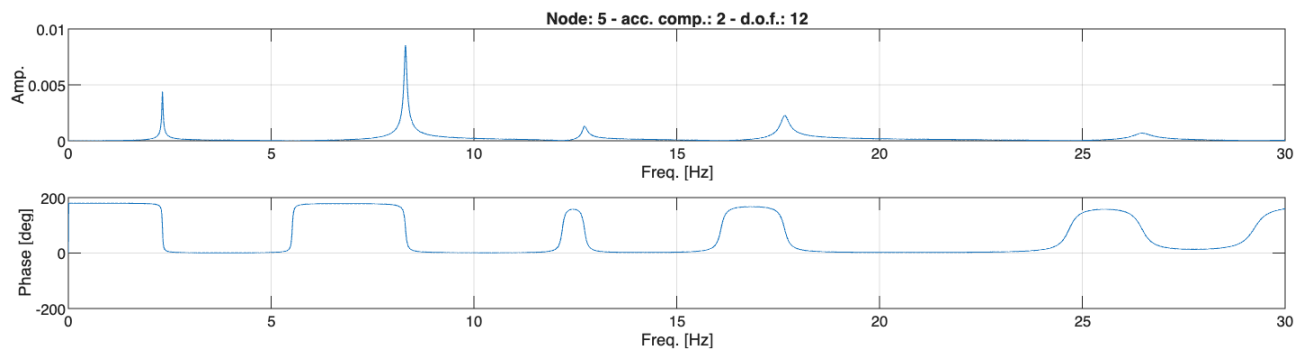


Figure 13: Vertical acceleration of point B produced by a vertical force on point B.

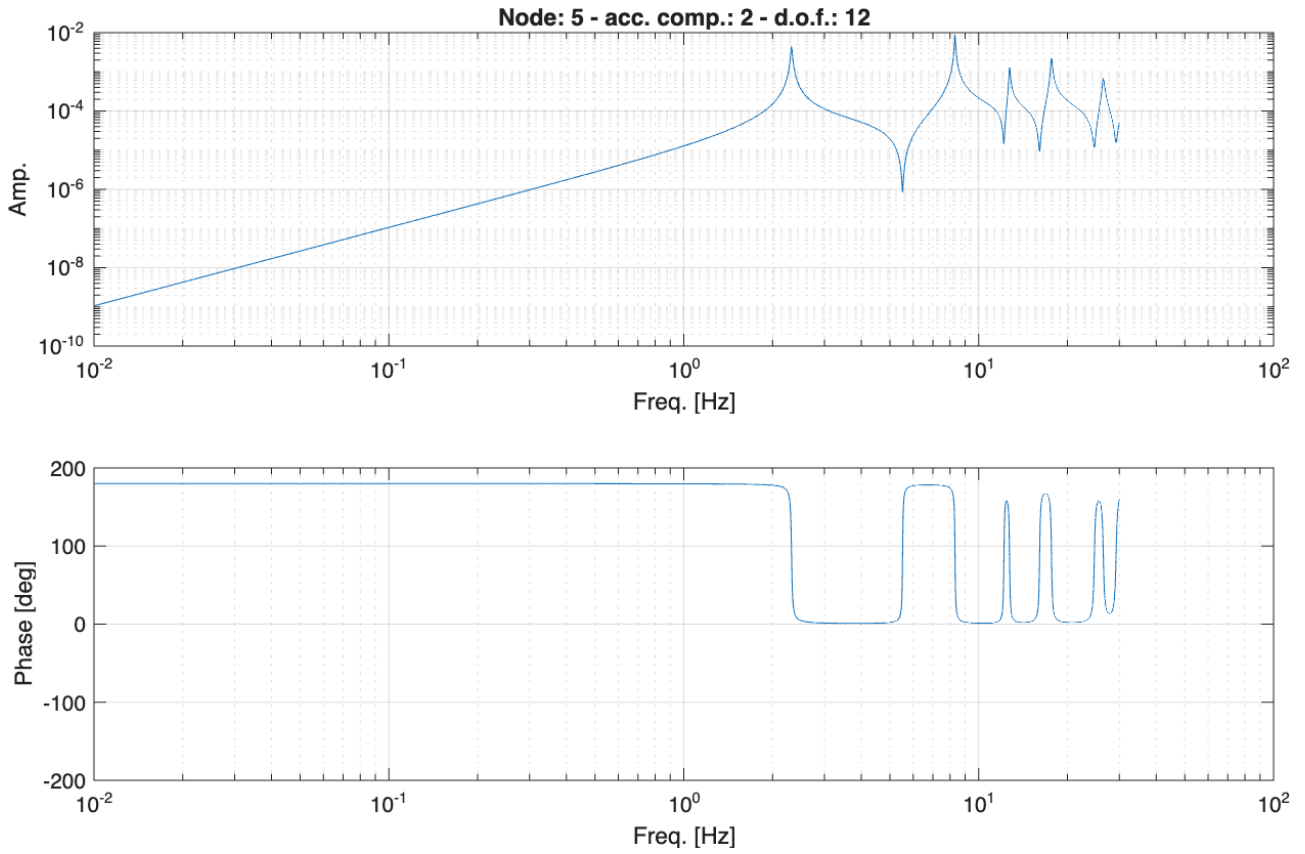


Figure 14: Vertical acceleration of point B produced by a vertical force on point B - Log scale

3.4 Practical considerations

Although the peak amplitude of 3.3×10^{-5} m/N appears numerically small, it represents a high dynamic compliance. Considering that operational loads on the bridge are in the order of hundreds of kilo-Newtons (10^5 N), a sustained excitation at the resonance frequency of 2.32 Hz would theoretically predict displacements in the order of meters. In practice, this implies that the Dynamic Amplification Factor at this frequency is extremely high, and even small harmonic loads could lead to exceeding the Serviceability Limit States (deflection limit $\approx L/600$) or causing rapid fatigue failure.

4 Point 4

4.1 The spectrum of the input displacements yO1, yO2;

This section addresses the dynamic response of the bridge under seismic excitation, theoretically modeled as a multi-support base excitation problem where the ground motion is imposed as vertical displacements at supports O1 and O2. In this framework, the supports are not fixed but move according to specific time histories. The equation of motion is therefore solved by partitioning the degrees of freedom into free structural nodes and constrained support nodes, effectively translating the ground displacements into equivalent forcing functions acting on the structure.

The procedure begins with the processing of the seismic input data provided in the `seismic_displ.txt` file.

- Time Domain Acquisition: Extracting the time histories of the vertical displacements Y O1 and Y O2 from the provided three-column data file (Time, O1, O2).
- Frequency Domain Transformation: Applying a Fast Fourier Transform (FFT) to these time histories to compute the input spectra. This transformation allows us to visualize the energy distribution of the earthquake across different frequencies, identifying which harmonic components will most strongly excite the bridge within the analyzed range of 0–30 Hz.

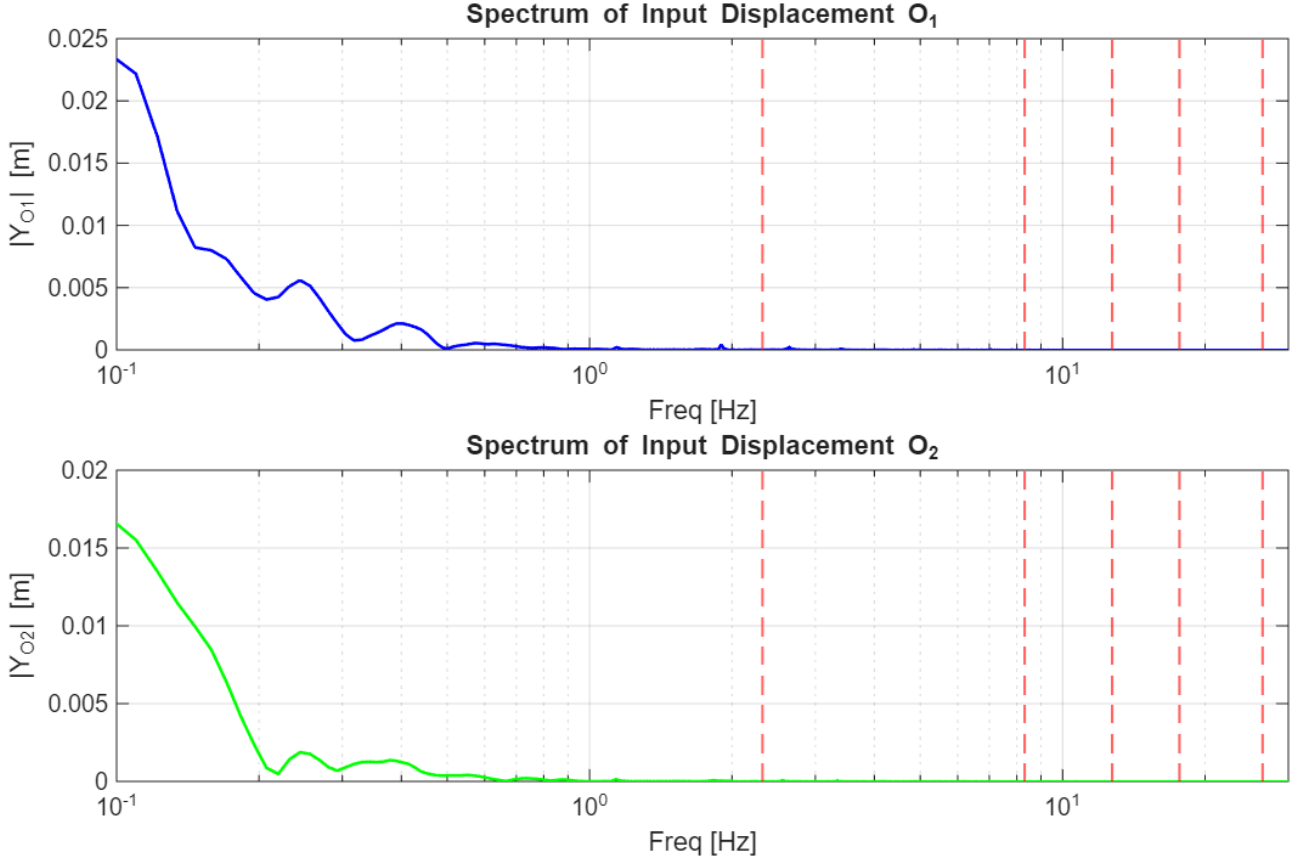


Figure 15: The spectrum of the input displacements y_{O1} , y_{O2}

Results analysis and comments To correctly interpret the seismic input and the structural response in the frequency domain, two specific post-processing techniques were applied to the raw Fast Fourier Transform (FFT) results: the use of a logarithmic scale for visualization and the conversion to a Single-Sided Spectrum.

- The computed spectrum exhibits a high concentration of energy in the very low-frequency range (near 0 Hz). This result is qualitatively correct and consistent with the physical nature of the input signal observed in the time domain. Looking at the time history plots of the ground displacements Y_{O1} and Y_{O2} , the signal is characterized by slow, broad oscillations rather than rapid vibrations. Visually, the dominant waves have large periods T spanning several seconds. Since frequency is inversely proportional to the period, these long durations mathematically correspond to frequencies well below 1 Hz ($0.1 \div 0.2$ Hz), which is confirmed by the obtained figures. Furthermore, seismic displacement is derived from the double integration of acceleration, an operation that scales spectral amplitude by $1/\omega^2$, acting as a strong low-pass filter. To properly visualize this dominant low-frequency content a semi-logarithmic scale was adopted. This allows for a clear distinction of peaks in the range of interest $0.1 \div 30$ Hz that would otherwise be compressed near the origin in a linear plot.
- The standard FFT algorithm produces a "Double-Sided" spectrum, distributing the signal's total energy equally between positive frequencies and negative frequencies. Since negative frequencies have no physical meaning in this structural analysis, we adopt the Single-Sided Spectrum representation. In compliance with the Nyquist-Shannon sampling theorem, the discrete signal is valid only up to the Nyquist frequency. Consequently, the spectrum is truncated at f_{Nyq} . To satisfy Parseval's theorem and preserve the total energy of the physical signal, the amplitude of the positive frequency components is multiplied by a factor of 2. This operation "folds" the energy from the discarded negative frequencies onto the positive ones, ensuring the plotted values represent the real physical amplitude of the displacement.

4.2 The spectrum of the vertical displacements of points A and B

The process begins by constructing the input matrix, X_{C_freq} , which represents the known ground motion in the frequency domain. The code takes the Fast Fourier Transforms (FFT) of the seismic time histories for supports O1 and O2 and maps them into the rows corresponding to the constrained vertical degrees of freedom. This step effectively translates the earthquake excitation from a time-varying signal into a frequency-dependent input that drives the system.

Once the inputs are prepared, the script enters a loop to solve the equation of motion for every frequency step. Inside this loop, the code calculates the dynamic stiffness matrix (A) for the free degrees of freedom and the effective force vector ($Qf0$). The effective force represents the seismic energy transferred from the moving supports to the structure, derived using the mass, damping, and stiffness coupling matrices (MFC,CFC,KFC). By solving the linear system $X0 = A^{-1} \cdot Qf0$ at each frequency, the script obtains the complex displacement vector for every node in the bridge, capturing the structural response across the entire frequency range.

Finally, the code extracts the specific results for Node A (mid-span) and Node B (located at $2L$) by identifying their corresponding row indices within the global displacement vector. Since the raw results are complex numbers, the script computes their magnitude to generate the response spectrum. These values are then plotted against frequency to visualize how the bridge oscillates, highlighting any peaks where the seismic input frequency aligns with the bridge's natural frequencies, which would indicate potential resonance conditions.

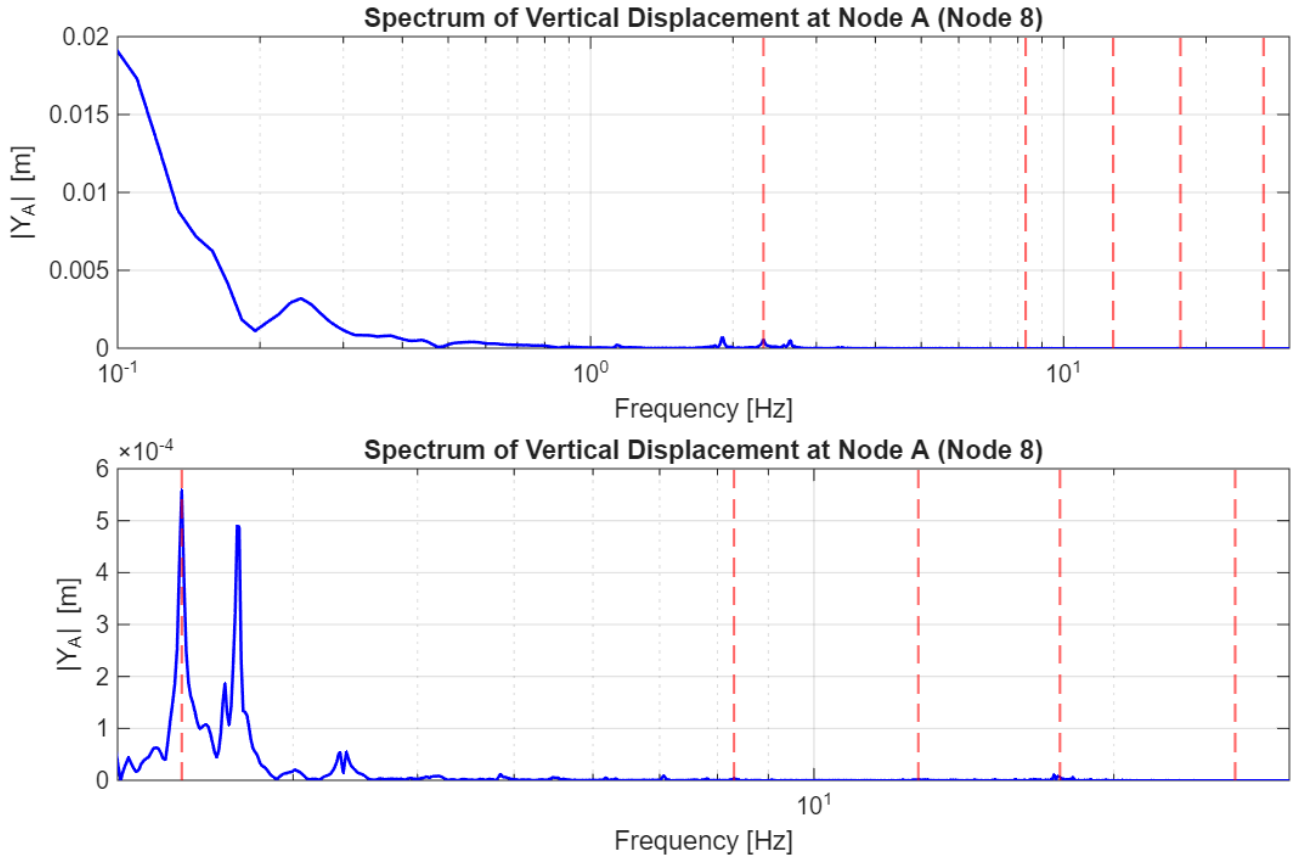


Figure 16: Spectrum of the displacement of point A

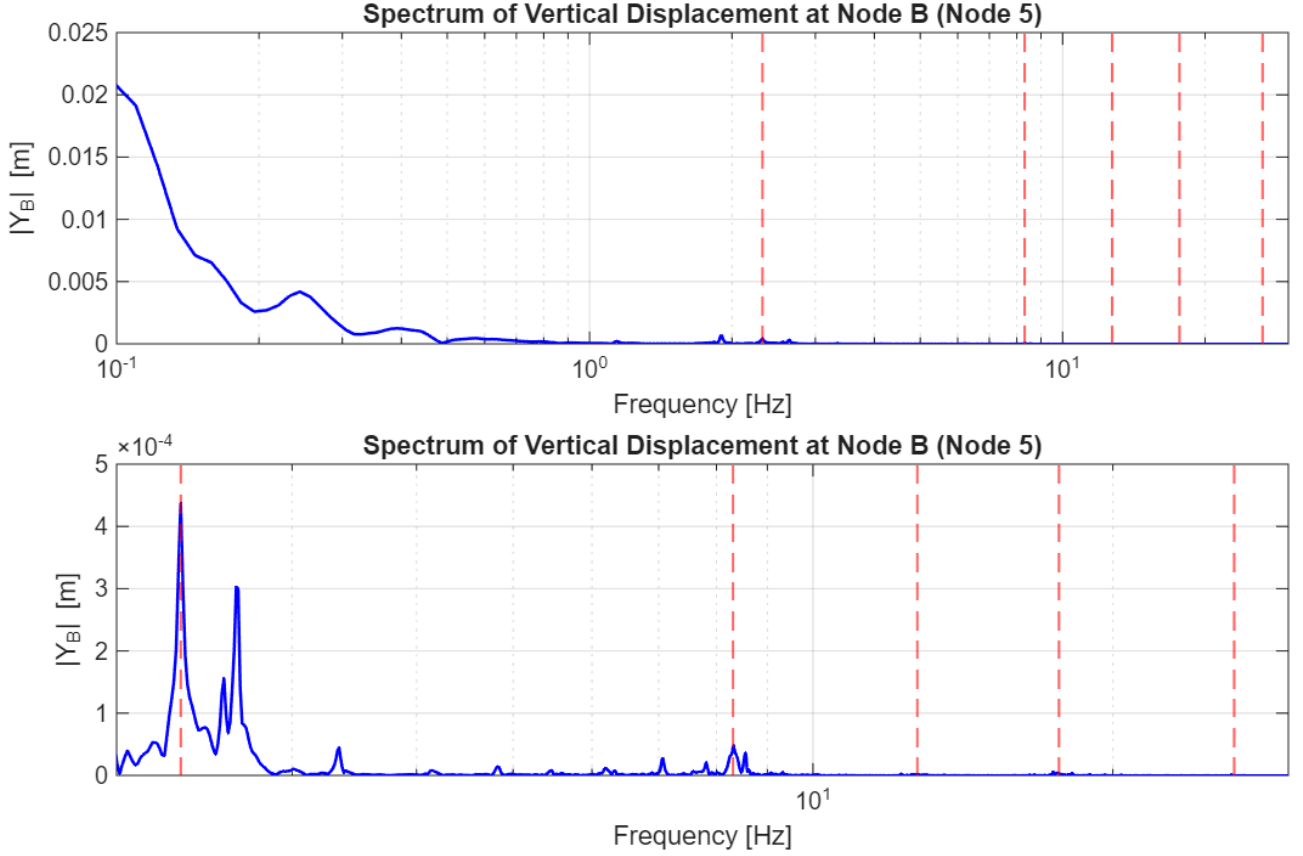


Figure 17: Spectrum of the displacement of point B

Results analysis and comments The computed displacement response spectra for both Node A and Node B exhibit a massive concentration of energy in the very low-frequency range ($f < 1$ Hz), while the structural resonance peaks (corresponding to the natural frequencies identified in Section 2) appear negligible or invisible. This behavior is physically consistent and it can be attributed to two main factors:

- **Spectral Content of the Input:** As analyzed in Section 4.1, the seismic displacement input is dominated by long-period components ($T > 5$ s). At these frequencies, the excitation frequency is much lower than the bridge's fundamental frequency.
- **Structural Behavior:** In this low-frequency regime, the structure behaves quasi-statically. The dynamic amplification factor is approximately unity, meaning the bridge moves "rigidly" with the ground motion without significant elastic deformation. The displacement of points A and B is simply the geometric interpolation of the settlements of constraints O_1 and O_2 . Conversely, at the resonance frequencies, although the dynamic amplification is high, the input displacement magnitude is extremely low (due to the $1/\omega^2$ decay of the seismic signal). The product of a high amplification factor and a negligible input results in a response peak that is orders of magnitude smaller than the low-frequency ground motion.

For this reason, the spectral results are presented in two complementary views: a full-scale plot to illustrate the dominant quasi-static contribution, and a detailed view with a rescaled y-axis to highlight the modal contributions within the resonance zone.

While both points A and B share the same dominant quasi-static profile, differences arise in the higher frequency range (resonance zone), dictated by the mode shapes:

- **Point A:** Located exactly at $L/2$, Point A corresponds to the antinode of the symmetric modes (Mode 1, Mode 3) but lies on a nodal point for antisymmetric modes (Mode 2 at 8.31 Hz). Consequently, even if the earthquake contained energy at 8.31 Hz, Point A would theoretically show zero response at this frequency. Its dynamic response is almost entirely governed by the fundamental Mode 1.
- **Point B** is not a nodal point for the first antisymmetric mode. Therefore, its spectrum contains contributions from a broader set of modes (both symmetric and antisymmetric). However, given the magnitude of the quasi-static component mentioned above.

4.3 The time histories of the vertical displacements of points A and B

This script converts the frequency-domain results back into the time domain using the Inverse Fast Fourier Transform (IFFT). By applying this transformation to the displacement spectrum calculated previously, the code reconstructs the actual vertical motion of the bridge ($y(t)$) throughout the earthquake's duration. It then extracts and plots the specific time histories for Node A and Node B, allowing you to visualize the transient response and identify the maximum displacements reached during the event.

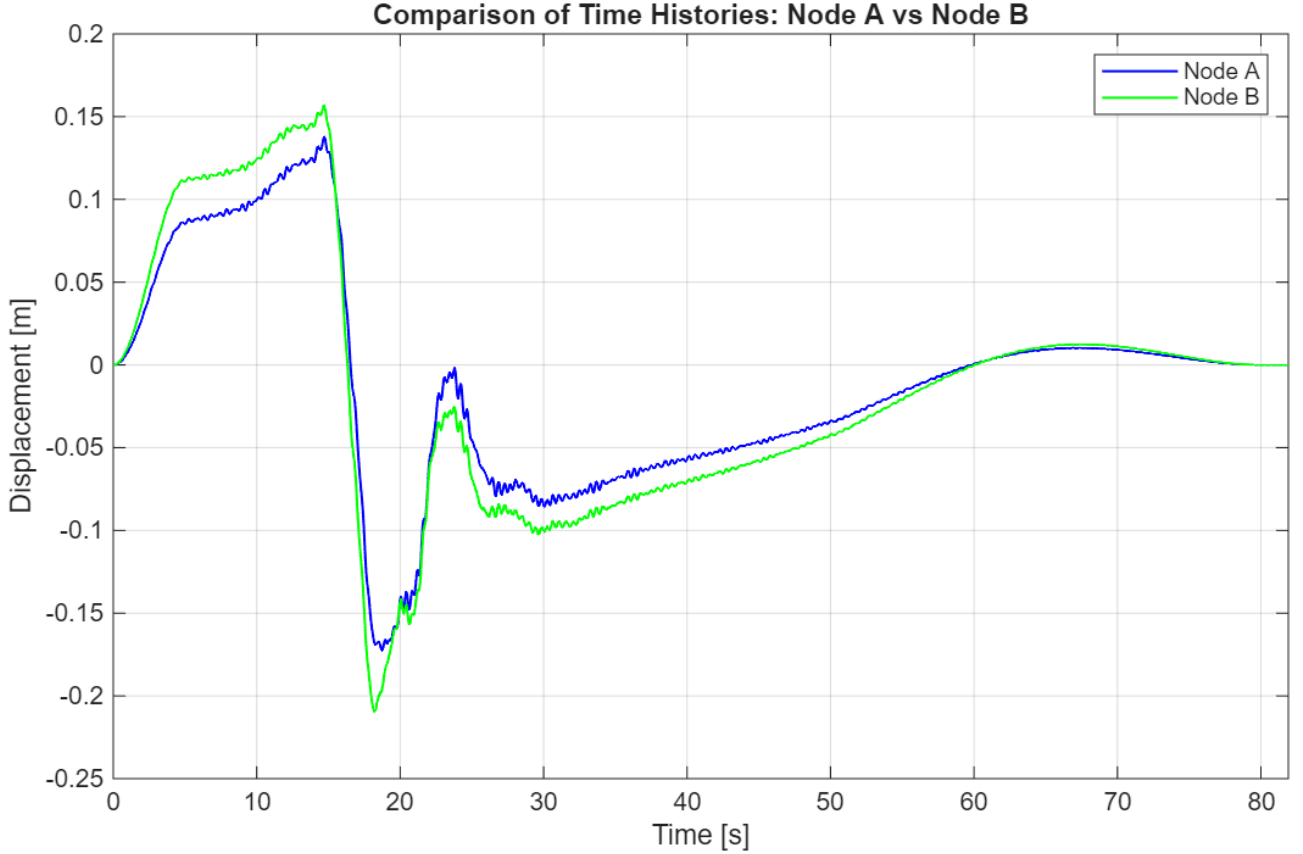


Figure 18: Time Histories of the displacement of point A and point B

Results analysis and comments The displacement time histories represent the instantaneous evolution of a point's position over time, due to a dynamic excitation. They are the direct visual representation of the structure's physical response.

In reading a Time History we need to be careful about the horizontal axis that represent time, the vertical axis that represents the response quantity, peaks and valleys indicating maximum and minimum response values, oscillations that reflect the dynamic nature of the system and its vibration modes and a decreasing amplitude over time that indicates the presence of damping.

Here the curves show significant activity only when the seismic input signal is active. The responses of Nodes A and Nodes B are very similar in shape and timing, indicating a global structure response dominated by the same vibration modes. This suggest that the bridge behaves coherently under the applied excitations. Node B generally exhibits a slightly larger displacement amplitudes than Node A, especially during the initial positive phase and first negative peak. This difference can be attributed to the relative position of Node B with respect to the applied external excitations and to the modal deformation shape of the structure.

Since the bridge is excited at both ends, the superposition of the dynamic waves propagating along the structure leads to a non-uniform displacement amplitudes at difference locations. Node B appears to be located closer to a region of higher modal participation, where the contribution of the dominant vibration modes is more pronounced. As a result the local dynamic response at Node B is slightly amplified compared to Node A. Both nodes show a gradual increase in positive displacement, corresponding to the build-up of deformation under the applied dynamic loads. The smooth growth indicates a predominantly low frequency response. A sharp transition from positive to negative displacement is observed for both nodes at approximately the same time instant. This behavior is associated with a change in the phase relationship between the two external excitations applied at the bridge extremities. When the two excitations become out of phase, the structural response undergoes a rapid redistribution of internal forces, leading to a reversal of the global bending deformation.

This result in a sudden shift of the displacement sign at both nodes, indicating a global dynamic effect rather than a local phenomenon. Node B reaches a more pronounced negative peak. After the main negative peak, both responses exhibit oscillations with decreasing amplitude, characteristic of a transient dynamic response governed by inertia and structural damping. The displacements gradually return toward zero, showing a damped vibration regime. The fact that the transition occurs almost simultaneously at Nodes A and B confirms that the response is dominated by the global modes of vibration of the bridge.

4.4 The spectrum and time histories of the vertical accelerations of points A and B

We first constructed the full two-sided frequency vector to match MATLAB's FFT ordering. We then calculated the acceleration in the frequency domain by multiplying the complex displacement spectrum by $-\omega^2$. Finally, by applying the Inverse Fast Fourier Transform (IFFT) to this result, we mathematically reconstructed the precise time-domain acceleration signals for Nodes A and B.

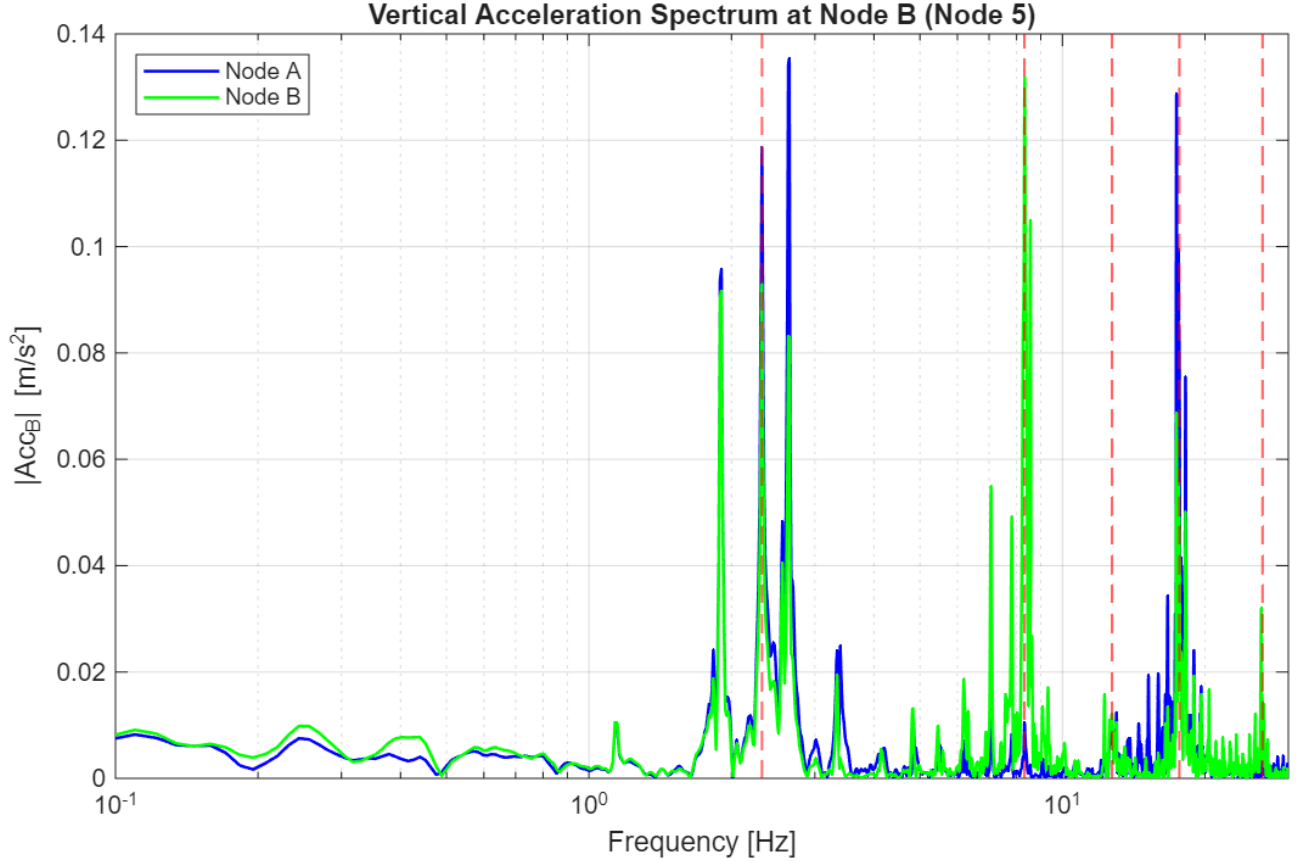


Figure 19: Spectrum of the acceleration of point A and B

Results analysis and comments The acceleration spectra exhibit a fundamentally different profile compared to the displacement spectra discussed in Section 4.2. While displacements were dominated by the low-frequency quasi-static component ($f < 1$ Hz), the acceleration response is dominated by the bridge's natural frequencies. This inversion is due to the kinematic relationship $A(\omega) = -\omega^2 X(\omega)$. The ω^2 factor acts as a potent high-pass filter: it suppresses the high-amplitude but slow ground motion (multiplying it by values $\ll 1$) and drastically amplifies the structural vibrations in the resonance range (multiplying by values $\gg 1$). Consequently, the "quasi-static mountain" visible in the displacement plots disappears, revealing the structure's dynamic signature.

The differences between the two points:

- Point A: The spectrum shows a distinct, high-amplitude peak at the fundamental frequency and subsequent symmetric modes. Crucially, the response at the second natural frequency is virtually absent (more than half of peak of point B). This is physically correct: Mode 2 is antisymmetric, possessing a node exactly at mid-span. The acceleration analysis confirms that Node A is "blind" to antisymmetric excitation.
- Point B is not a stationary point for the antisymmetric modes. Accordingly, its spectrum displays significant peaks at all identified natural frequencies. This confirms that general points on the deck are

subjected to a complex superposition of both symmetric and antisymmetric vibrations.

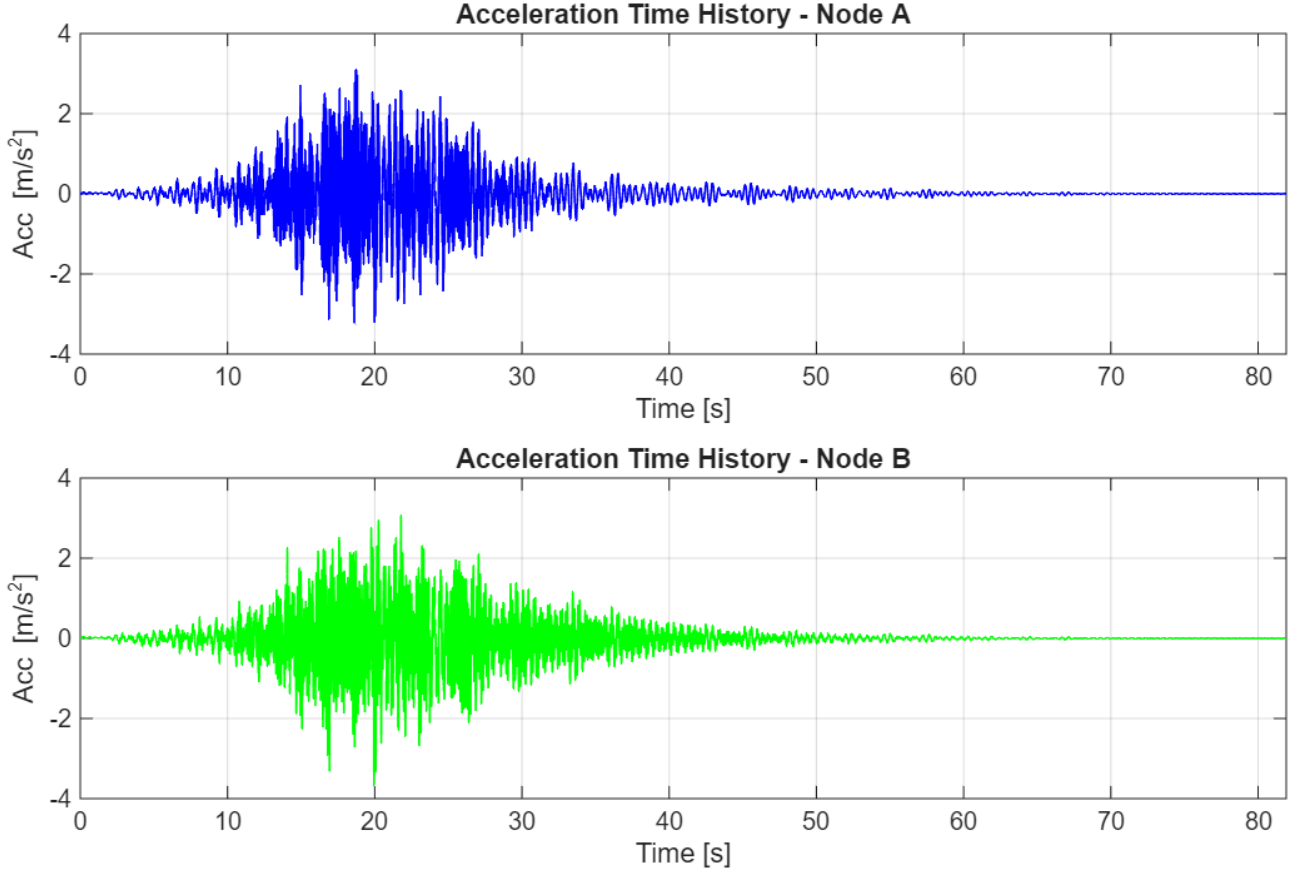


Figure 20: Time Histories of the vertical acceleration of point A and B

Results analysis and comments The time-domain representation of the vertical accelerations at Nodes A and B provides a direct visualization of the dynamic stress experienced by the structure during the seismic event.

The acceleration response appears as a much higher-frequency signal, characterized by rapid oscillations. The bridge acts as a mechanical band-pass filter: it suppresses the slow drift of the ground and selectively amplifies the components of the earthquake that align with its own natural frequencies. It can be seen that in the time domain, the bridge oscillates roughly 2.5 times per second – this corresponds to the first natural mode identified in Section 2. This indicates that, despite the broadband nature of the seismic acceleration input, the bridge primarily responds by vibrating in its fundamental mode shape. Higher frequency modes appear as smaller ripples superimposed on this main oscillation, while the 2nd mode is contributing only at Node B (as discussed in the spectral section) but is visually masked by the dominant lower-frequency amplitude.

Unlike displacement, where the maximum value was simply the ground movement, the acceleration peaks are generated by the elastic rebound of the structure. The fact that these oscillations persist and modulate in amplitude even as the strong ground motion varies, highlights the role of the damping ratio. The decay of these oscillations is governed by the structural damping matrix $[C]$, which dissipates the energy accumulated during the resonance cycles.

For a standard seismic event, peak structural accelerations are reasonably expected to be in the order of a fraction of the gravitational acceleration g - the computed peak values fall strictly within this expected physical range.

5 Point 5

5.1 Theoretical background

In this section the bridge is subjected to a moving train load. The train is idealised as an “infinite” sequence of identical axle loads P , travelling at constant speed V and spaced by a fixed distance d along the track. From the point of view of the structure, a generic point on the deck experiences a sequence of load pulses with constant

time spacing

$$\tau = \frac{d}{V}, \quad (33)$$

so that the resulting excitation is *periodic* in time with fundamental frequency

$$f_0 = \frac{1}{\tau} = \frac{V}{d}, \quad \Omega_0 = 2\pi f_0 = 2\pi \frac{V}{d}. \quad (34)$$

In modal coordinates, each mode i of the bridge behaves as a single-degree-of-freedom (SDOF) system with modal mass m_i , modal stiffness k_i , modal damping c_i and natural frequency

$$\omega_i = 2\pi f_i = \sqrt{\frac{k_i}{m_i}}, \quad (35)$$

governed by

$$m_i \ddot{q}_i(t) + c_i \dot{q}_i(t) + k_i q_i(t) = Q_i(t), \quad (36)$$

where $q_i(t)$ is the generalised coordinate and $Q_i(t)$ is the generalised force associated with the moving train load.

Because the excitation is periodic with fundamental frequency f_0 , the generalised force $Q_i(t)$ can be represented as a Fourier series,

$$Q_i(t) = \sum_{k=1}^{\infty} Q_{ik} \cos(k\Omega_0 t + \Psi_{ik}), \quad (37)$$

where $k = 1, 2, \dots$ is the harmonic order, Q_{ik} is the amplitude of the k -th harmonic of the generalised force on mode i and Ψ_{ik} is a phase angle. The train therefore excites each mode through a set of harmonics with angular frequencies

$$\Omega_k = k\Omega_0 = 2\pi k \frac{V}{d} \iff f_k = kf_0 = k \frac{V}{d}. \quad (38)$$

For a given mode i , resonance may occur when one of these excitation frequencies is close to the natural frequency f_i , that is when

$$kf_0 \approx f_i \iff k \frac{V}{d} \approx f_i. \quad (39)$$

This condition can be rewritten in terms of the corresponding *resonant train speed* $V_{i,k}$ for mode i and harmonic k :

$$V_{i,k} \approx \frac{f_i d}{k}. \quad (40)$$

Equation (40) provides, for each mode and each harmonic, a candidate train speed at which that mode might be strongly excited by the moving load.

5.2 Identification of candidate resonant speeds

The starting point of the analysis is the reduced model in terms of the free degrees of freedom, obtained by partitioning the global mass, damping and stiffness matrices and enforcing the boundary conditions at nodes O1 and O2 (hinge and cart). From the reduced mass and stiffness matrices M_{FF} and K_{FF} , the following generalised eigenvalue problem is solved:

$$K_{FF} \phi_i = \lambda_i M_{FF} \phi_i, \quad (41)$$

yielding the natural frequencies and mode shapes of the bridge:

$$\omega_i = \sqrt{\lambda_i}, \quad f_i = \frac{\omega_i}{2\pi}, \quad i = 1, 2, \dots \quad (42)$$

Only the modes with $0 < f_i \leq 30$ Hz are retained, consistently with the frequency range for which the finite element model has been designed.

For the moving-load analysis, the axle spacing is fixed at

$$d = 26 \text{ m},$$

as specified in the assignment. The resonance condition (40) is then applied to each mode i and to the first few harmonics $k = 1, 2, 3$, which are expected to carry the largest fraction of the excitation energy. Higher-order harmonics ($k > 3$) are neglected, because for a train of axle loads the Fourier coefficients Q_{ik} of the generalised force decay rapidly with k ; consequently, the contribution of higher harmonics to the modal response is much smaller than that of the fundamental and the first few harmonics.

For each mode i and harmonic order $k = 1, 2, 3$, the corresponding resonant speed is computed from (40) as

$$V_{i,k} = \frac{f_i d}{k} \text{ [m/s]}, \quad V_{i,k}^{(\text{km/h})} = 3.6 V_{i,k} \text{ [km/h]}. \quad (43)$$

Only speeds in a realistic operational range for railway traffic are considered; here, an upper bound

$$0 < V_{i,k}^{(\text{km/h})} \leq 400 \text{ km/h}$$

is imposed. All combinations (i, k) satisfying this condition are collected in a table, which reports, for each mode and harmonic, the associated candidate resonant train speed. The combinations are first sorted by increasing speed to provide an overview of the modes that may be resonantly excited in different speed ranges. An example of such a map, showing the pairs $(V_{i,k}, f_i)$ for all admissible combinations, is reported in Figure 21.

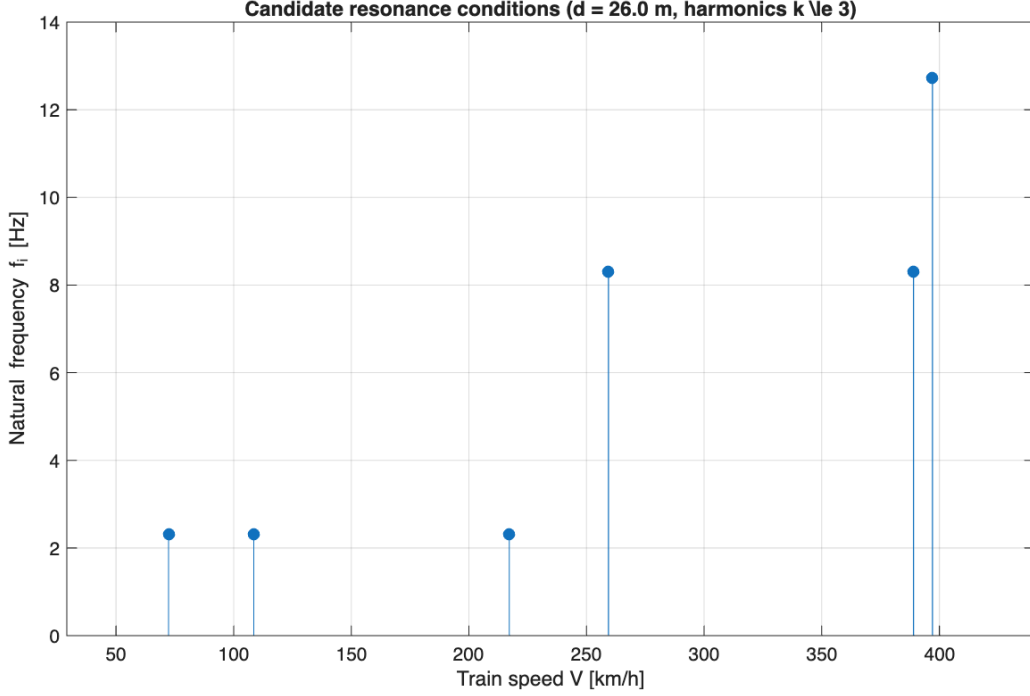


Figure 21: Candidate resonance conditions for the bridge: natural frequencies f_i versus train speed V for the first three harmonics ($k \leq 3$).

5.3 Modal damping and dynamic amplification

The purely kinematic condition (40) identifies the speeds at which a given mode is *in principle* prone to resonance, but it does not yet quantify how large the actual response will be. The severity of resonance depends crucially on the modal damping ratios ζ_i .

The reduced damping matrix C_{FF} (exported from the FEM code together with M_{FF} and K_{FF}) is projected onto the modal basis to obtain the modal mass, stiffness and damping coefficients:

$$m_i = \phi_i^T M_{FF} \phi_i, \quad k_i = \phi_i^T K_{FF} \phi_i, \quad c_i = \phi_i^T C_{FF} \phi_i. \quad (44)$$

The modal damping ratio of mode i is then given by the standard SDOF relation

$$\zeta_i = \frac{c_i}{2\sqrt{k_i m_i}} = \frac{c_i}{2m_i \omega_i}. \quad (45)$$

For a single mode subjected to a harmonic generalised force at frequency Ω_k , the steady-state response amplitude is proportional to the modulus of the frequency response function

$$H_i(\Omega_k) = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta_i r)^2}}, \quad r = \frac{\Omega_k}{\omega_i}, \quad (46)$$

which represents the ratio between the dynamic and static responses under a harmonic excitation. Assuming that mode i is excited by the k -th harmonic of the train loading, the corresponding angular frequency is

$$\Omega_k = 2\pi k \frac{V}{d}, \quad (47)$$

so that the dynamic amplification factor $H_{i,k}$ can be evaluated for each candidate resonant speed $V_{i,k}$ identified previously.

For every combination (i, k) retained in the speed map, the corresponding damping ratio ζ_i and amplification factor $H_{i,k}$ are computed and added to the table of candidate resonances. The table is then re-sorted in decreasing order of $H_{i,k}$, in order to highlight the potentially most critical cases (“worst-case” combinations) in terms of *intrinsic* dynamic amplification of mode i at the excitation frequency of the k -th harmonic. An example of the resulting table is shown in Table 2, where each row contains: the mode index i , its natural frequency f_i , the harmonic order k , the associated train speed $V_{i,k}$ and the amplification factor $H_{i,k}$.

It is important to remark that, by construction, the resonant speeds $V_{i,k} = f_i d/k$ are chosen so that the k -th harmonic of the moving load is exactly tuned to the natural frequency of mode i . In fact, with $f_0 = V_{i,k}/d = f_i/k$ one has

$$\Omega_0 = 2\pi f_0 = \frac{2\pi f_i}{k}, \quad \Omega_k = k\Omega_0 = 2\pi f_i = \omega_i,$$

so that the frequency ratio in the SDOF response function, $r = \Omega_k/\omega_i$, is equal to one for all harmonics k associated with the same mode i . As a consequence, the amplification factor reduces to the peak value

$$H_{i,k} = H_i(\Omega_k) = H_i(\omega_i) \approx \frac{1}{2\zeta_i},$$

which depends only on the modal damping ratio ζ_i and is *independent* of the harmonic order k for a given mode. Sorting the combinations by $H_{i,k}$ therefore provides information on how strongly each mode can resonate in principle (given its damping), but it does not distinguish between the effects of different harmonics of the moving load on that mode.

In practice, the actual severity of each resonance also depends on the amplitude of the k -th harmonic in the generalised force, described by the Fourier coefficients Q_{ik} , which typically decrease rapidly with increasing k for a train of axle loads. Low-order harmonics (especially $k = 1$) are therefore associated with much stronger excitation than higher-order harmonics, even if the intrinsic dynamic amplification $H_{i,k}$ is the same. Moreover, different modes exhibit different participation patterns in the global bridge response. For these reasons, the combinations with the largest $H_{i,k}$ are not automatically the most dangerous in practice; this aspect is discussed in more detail in the following subsection, where the worst-case scenario is selected by jointly considering harmonic order and modal participation.

Table 2: Candidate resonant train speeds for the first three harmonics and modes with $f_i \leq 30$ Hz. For each combination (i, k) the table reports the natural frequency f_i , the harmonic order k , the corresponding train speed $V_{i,k}$ and the dynamic amplification factor $H_{i,k}$.

Mode i	f_i [Hz]	k	$V_{i,k}$ [km/h]	$H_{i,k}$ [-]
2	8.312	3	259.3	132.06
2	8.312	2	389.0	132.06
3	12.726	3	397.1	110.22
1	2.320	3	72.4	86.19
1	2.320	2	108.6	86.19
1	2.320	1	217.1	86.19

5.4 Selection and discussion of the worst-case scenario

The amplification factors $H_{i,k}$ quantify how strongly each mode is excited by the moving load at the corresponding resonant speed $V_{i,k}$, taking into account the actual modal damping. However, the severity of a resonance does not depend solely on $H_{i,k}$. Two additional aspects must be considered:

- the *harmonic order* k : the Fourier coefficients Q_{ik} of the generalised force decrease rapidly with k , so that the excitation associated with higher harmonics ($k \geq 3$) is much weaker than that associated with the fundamental ($k = 1$); therefore, cases with large $H_{i,k}$ but high harmonic order may be less critical in practice;
- the *modal participation* in the global response: lower modes, and in particular the fundamental bending mode, usually exhibit larger vertical displacements over wide portions of the span and therefore contribute more significantly to the overall structural response under moving loads.

In the present case, the analysis of Table 2 shows that:

- several combinations (i, k) exhibit relatively large amplification factors $H_{i,k}$, especially for lightly damped modes;
- among these, the combination corresponding to the *first bending mode* in resonance with the *first harmonic* of the moving load (i.e. $k = 1$) occurs at a train speed of approximately

$$V_{\text{res},1,1} \approx 217 \text{ km/h},$$

which lies within a realistic operating range for modern trains;

- higher-order modes in resonance with the second or third harmonic may exhibit comparable or even slightly larger $H_{i,k}$ values, but they are associated with higher harmonic orders $k = 2$ or $k = 3$ (and therefore smaller excitation amplitudes).

On this basis, the combination “mode 1, $k = 1$, $V \approx 217 \text{ km/h}$ ” is identified as the *worst-case scenario* for the bridge under the moving train load. For this critical case, the variation of the amplification factor as a function of the train speed is further investigated by evaluating

$$H_1(V) = \frac{1}{\sqrt{(1 - r(V)^2)^2 + (2\zeta_1 r(V))^2}}, \quad r(V) = \frac{\Omega_1(V)}{\omega_1} = \frac{2\pi V}{d}, \quad (48)$$

in a neighborhood of the resonant speed $V_{\text{res},1,1}$. The resulting curve $H_1(V)$ is plotted in Figure 22, together with the theoretical peak value $H_{\text{max}} \approx 1/(2\zeta_1)$ for a perfectly tuned resonance and with a vertical line marking the resonant speed.

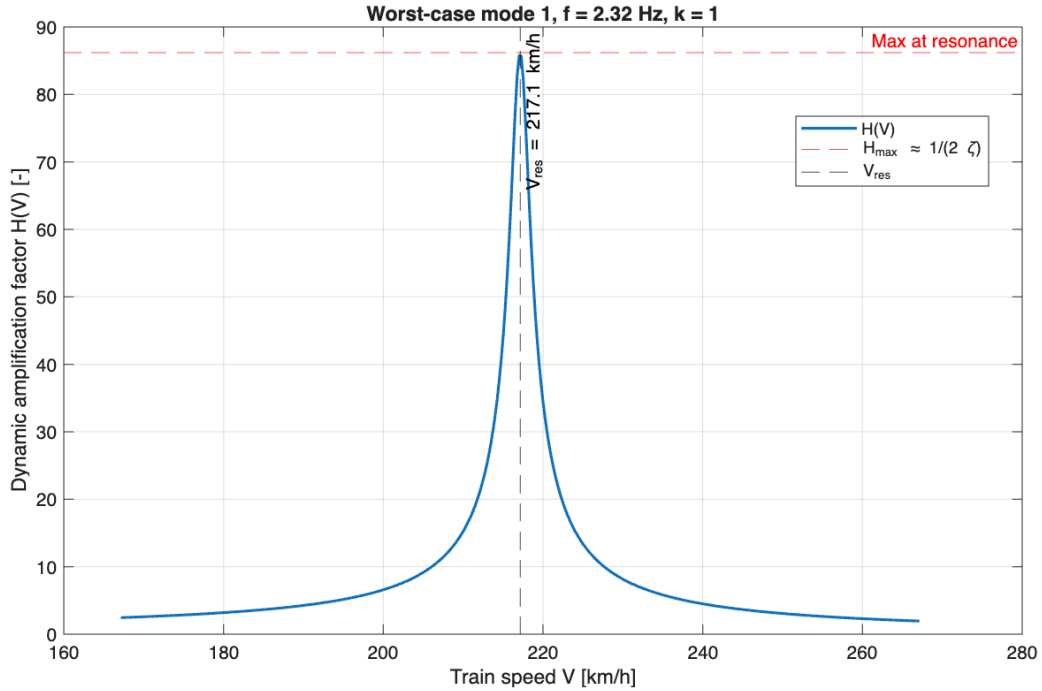


Figure 22: Dynamic amplification factor $H_1(V)$ of the first bending mode as a function of the train speed V in the vicinity of the critical speed $V_{\text{res},1,1} \approx 217 \text{ km/h}$. The horizontal dashed line indicates the theoretical maximum $H_{\text{max}} \approx 1/(2\zeta_1)$, while the vertical line marks the resonant speed.

The plot confirms that the amplification factor exhibits a pronounced peak around $V \approx 217 \text{ km/h}$, whose height is limited by the modal damping ratio ζ_1 . As the train speed moves away from this value, the amplification rapidly decreases, so that the response remains moderate over a wide range of speeds. Taken together, the analysis of candidate resonant speeds, modal damping ratios and amplification factors indicates that the first bending mode at approximately 217 km/h represents the most critical configuration for the bridge under the considered moving load model, while other combinations with larger $H_{i,k}$ but higher harmonic order or higher mode index are less worrying in practice because of the smaller excitation amplitudes and reduced participation at the monitoring points.

5.5 Graphic analysis of first mode and comparisons

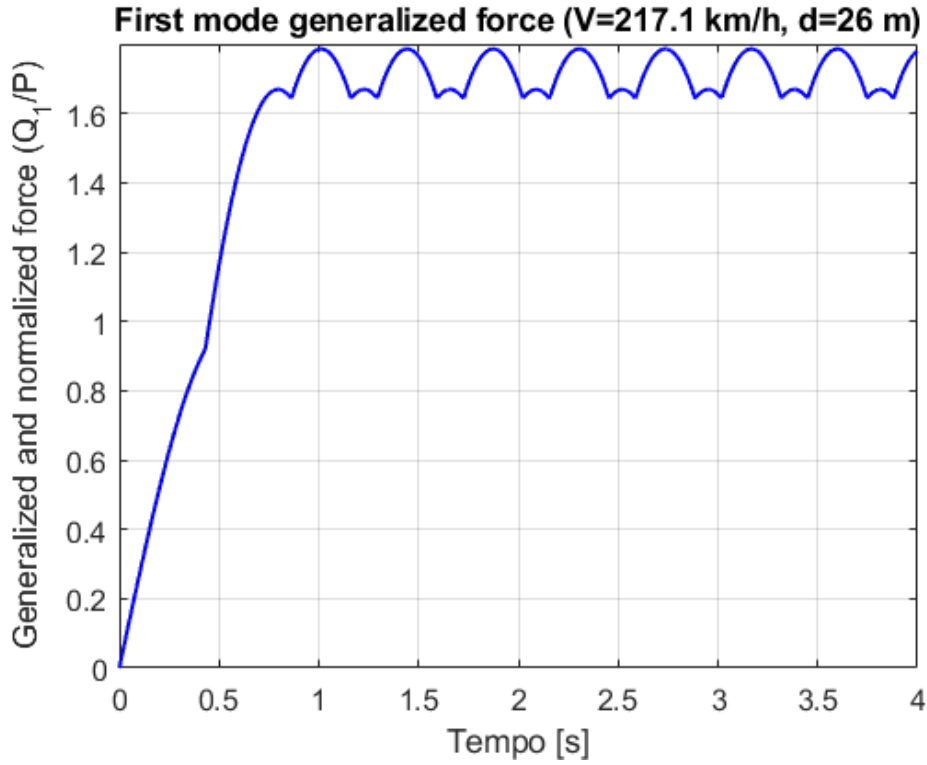


Figure 23: First mode when a train passes with 217 km/h passes

Figure 23 depicts the response of the first vibration mode. It is evident that the moving load contributions superimpose constructively. Following a transient initial phase, the system develops a steady-state response. At a velocity of 217.1 km/h, the excitation frequency matches the system's first natural frequency ($f_1 \approx 2.3$ Hz), triggering a resonance phenomenon. Consequently, the amplitude of oscillation diverges over time, leading to a condition of structural instability (SLU).

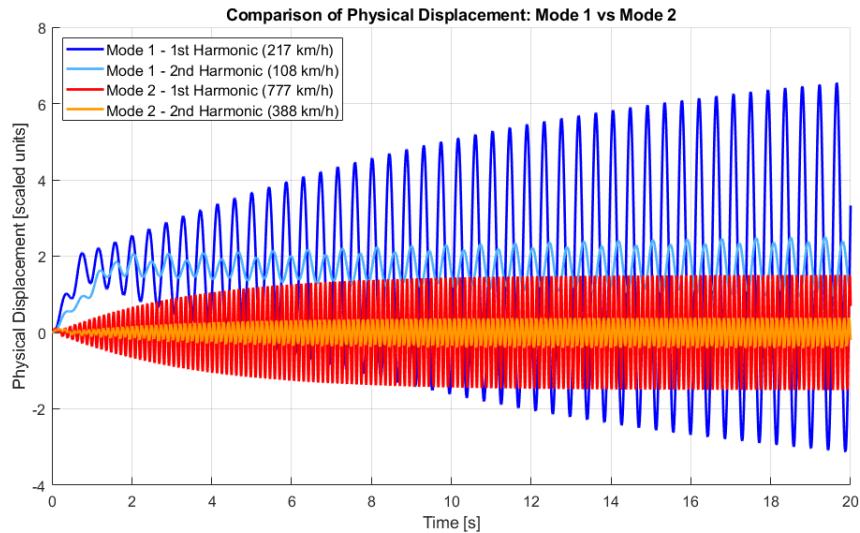


Figure 24: First mode when a train passes with 217 km/h passes

Extending the analysis to a broader spectrum of velocities allows for a comprehensive understanding of the structural behavior. As illustrated in Figure 24, the dynamic amplification factor is maximized at the critical velocity of 217 km/h (Fundamental Resonance, $k = 1$). The analysis highlights that the second harmonic of the first mode ($k = 2$, corresponding to 108 km/h) also presents a significant risk. Although the displacements

are smaller compared to the fundamental resonance, which would likely cause immediate structural failure, the second harmonic generates substantial cyclic stresses. This could drastically reduce the bridge's service life due to fatigue damage. Regarding the second mode ($f_2 \approx 8.3$ Hz), the fundamental resonance speed (777 km/h) is physically unattainable. Even the second harmonic speed (388 km/h) represents an unrealistic scenario. Furthermore, the graph demonstrates that the displacement amplitudes associated with the second mode are negligible compared to the first mode. Therefore, the primary design constraint is the resonance of the first mode at 217 km/h. From a physical perspective, the dominance of the first mode is attributable to its shape and stiffness.

6 Point 6

Introduction Define a structural modification capable of reducing the maximum vertical acceleration at Point B by at least 15% under a specific dynamic load, while respecting a strict mass constraint (maximum increase of +5% of the total mass) and without introducing external constraints.

GOAL

$$acc_{original,max} = 8.55 \cdot 10^{-3} m/s \implies acc_{goal,max} \cdot 0.85 = 7.27 \cdot 10^{-3} m/s \implies acc_{new} \leq acc_{goal,max}$$

LIMIT

$$m_{original} = 22698 kg \implies m_{lim} = m_{original} \cdot 1.05 = 23833 kg \implies m_{new} \leq m_{lim}$$

In order to achieve this goal, three main engineering strategies were preliminarily evaluated:

1. **Material Substitution:** Replacing the structural steel with alternative materials to improve the stiffness-to-mass ratio (E/ρ).
2. **Geometric/Topological Modification:** Adding new structural members to change the load path.
3. **Cross-Section Optimization:** Modifying the geometric properties (Area A and Moment of Inertia I) of the existing members to redistribute mass and stiffness efficiently.
4. **Reduction of Vertical Acceleration via Supplementary Damping**

6.1 Not feasible material substitution:

The peak acceleration in resonance conditions is inversely proportional to the system's damping. Since the bridge utilizes a Rayleigh damping model ($[C] = \alpha[M] + \beta[K]$), the most effective way to increase damping—and thus reduce acceleration—without adding significant mass is to drastically increase the global stiffness $[K]$.

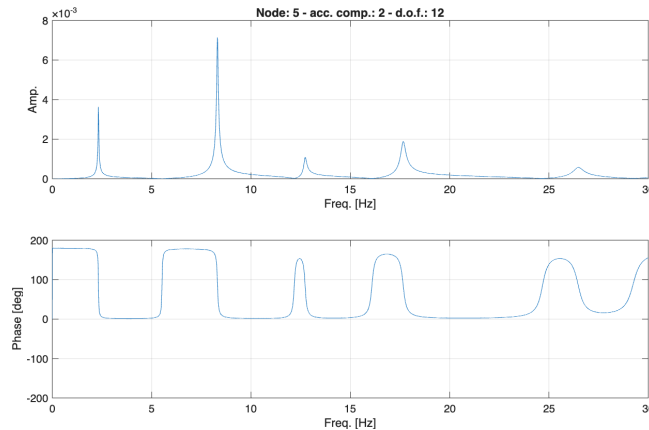


Figure 25: Enter Caption

An analysis of available engineering materials leads to the conclusion that changing the material is not a viable strategy for this specific application, for the following reasons:

- *Steel Alloys and Stiffness:* Common engineering intuition might suggest using "stronger" steel. However, while high-strength steel alloys offer higher yield strength (σ_y), they all share virtually the same Young's Modulus ($E \approx 210$ GPa). Since the problem is governed by deformation and not by failure, changing the steel grade provides zero benefit in terms of vibration reduction.

- *Alternative Materials:*

Aluminum offers a significant mass reduction ($\rho \approx 2700 \text{ kg/m}^3$), but its stiffness is roughly one-third that of steel ($E \approx 70 \text{ GPa}$). To maintain the same global stiffness, the cross-sections would need to be tripled in size, negating the weight advantage.

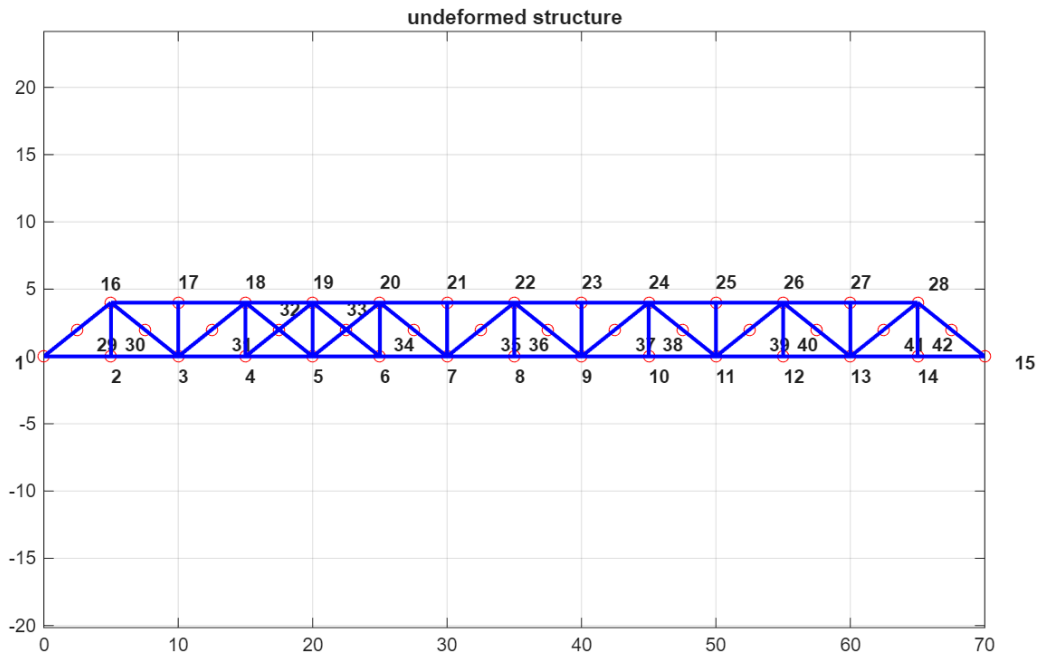
Titanium has a similar specific stiffness (E/ρ) to steel but comes with prohibitive costs and no significant dynamic advantage for this specific topology.

Carbon Fiber Reinforced Polymers (CFRP) theoretically offer the ideal solution, with a stiffness-to-weight ratio far superior to steel. However, the use of CFRP would require a complete redesign of the joint connections and introduces anisotropy issues that are complex to model in a 1D beam element formulation. Furthermore, the prohibitive cost makes it unrealistic for a standard civil infrastructure application.

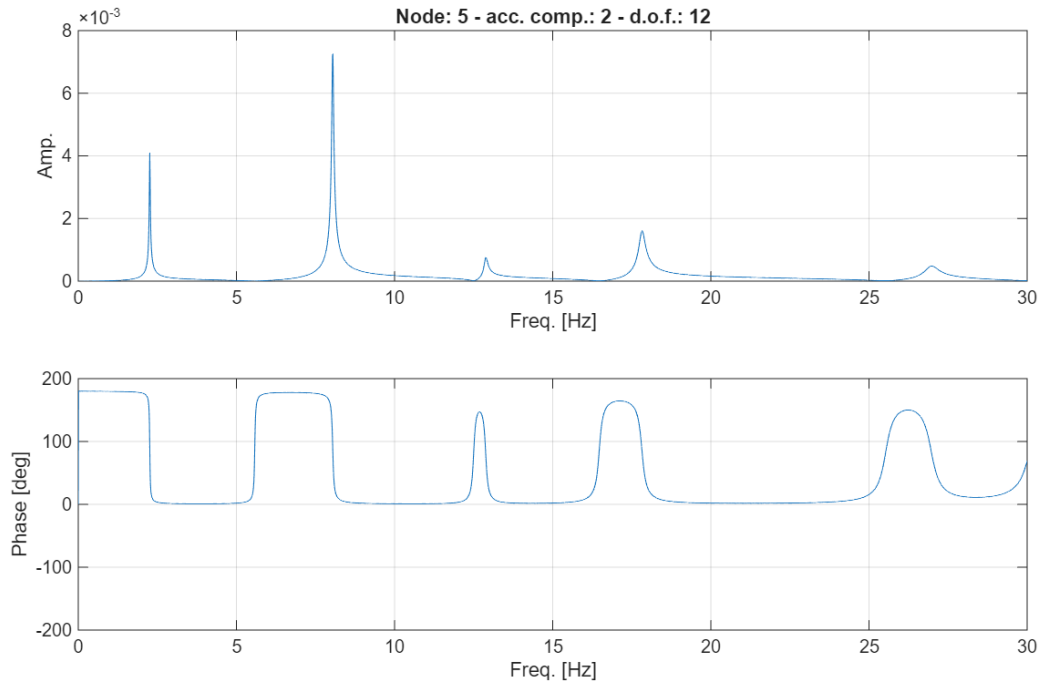
6.2 Geometric/Topological Modifications:

6.2.1 First case

The concept behind this structural choice is simplicity and ease of implementation. This is a purely additive solution: instead of modifying or removing any part of the current structure, we are simply introducing new elements to stiffen the area of node 5. The added beams have the same section as the diagonal beams



Total mass: 23823 kg (+4,975%)



Max acceleration: $7.25 \cdot 10^{-3} m/s(-15, 2\%)$

6.2.2 Second case

The new mass of 23720 that respect the +5% increasing of weight limit and its obtained using IPE240 for all the new beams e changing the beam 7-35-22 and 9-36-22 from HEA300 to IPE240

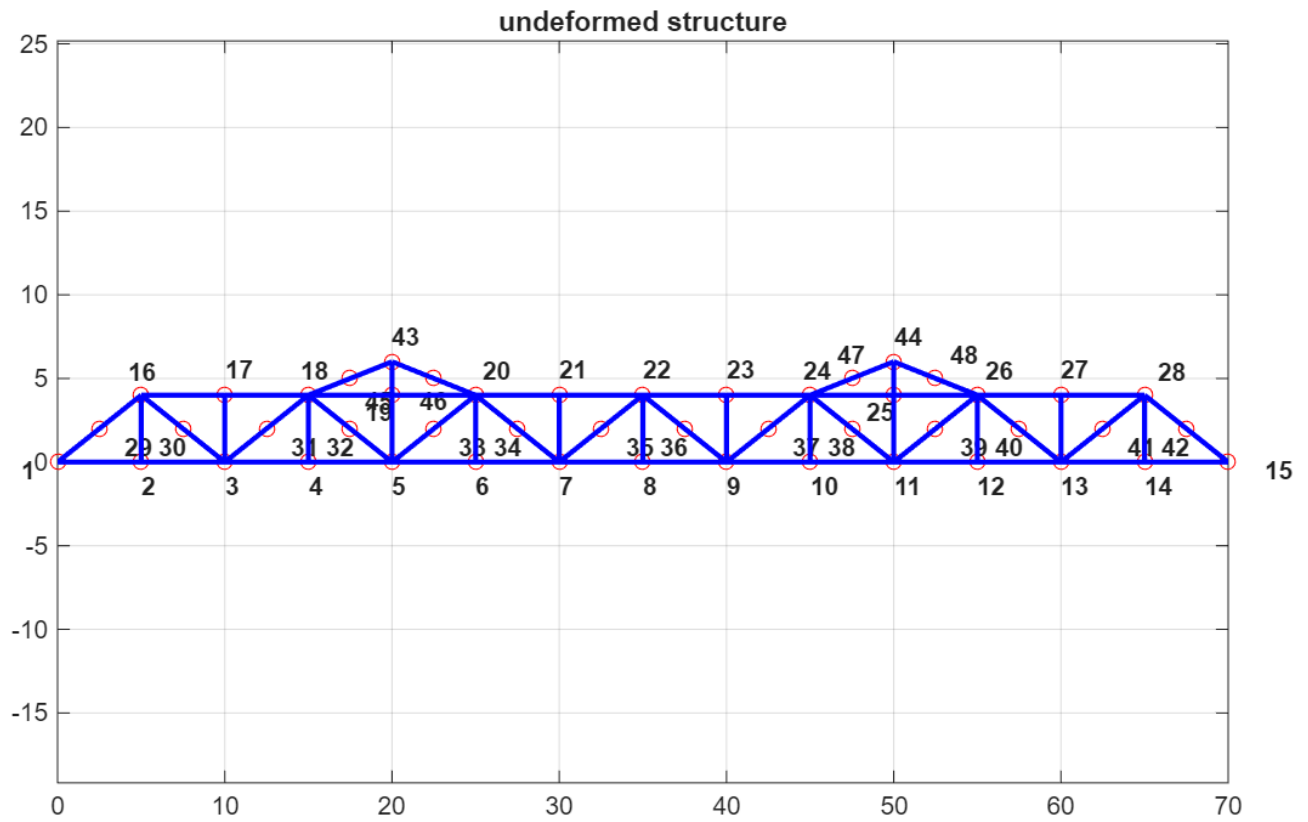


Figure 26: Undeformed structure of a version 2 of the bridge

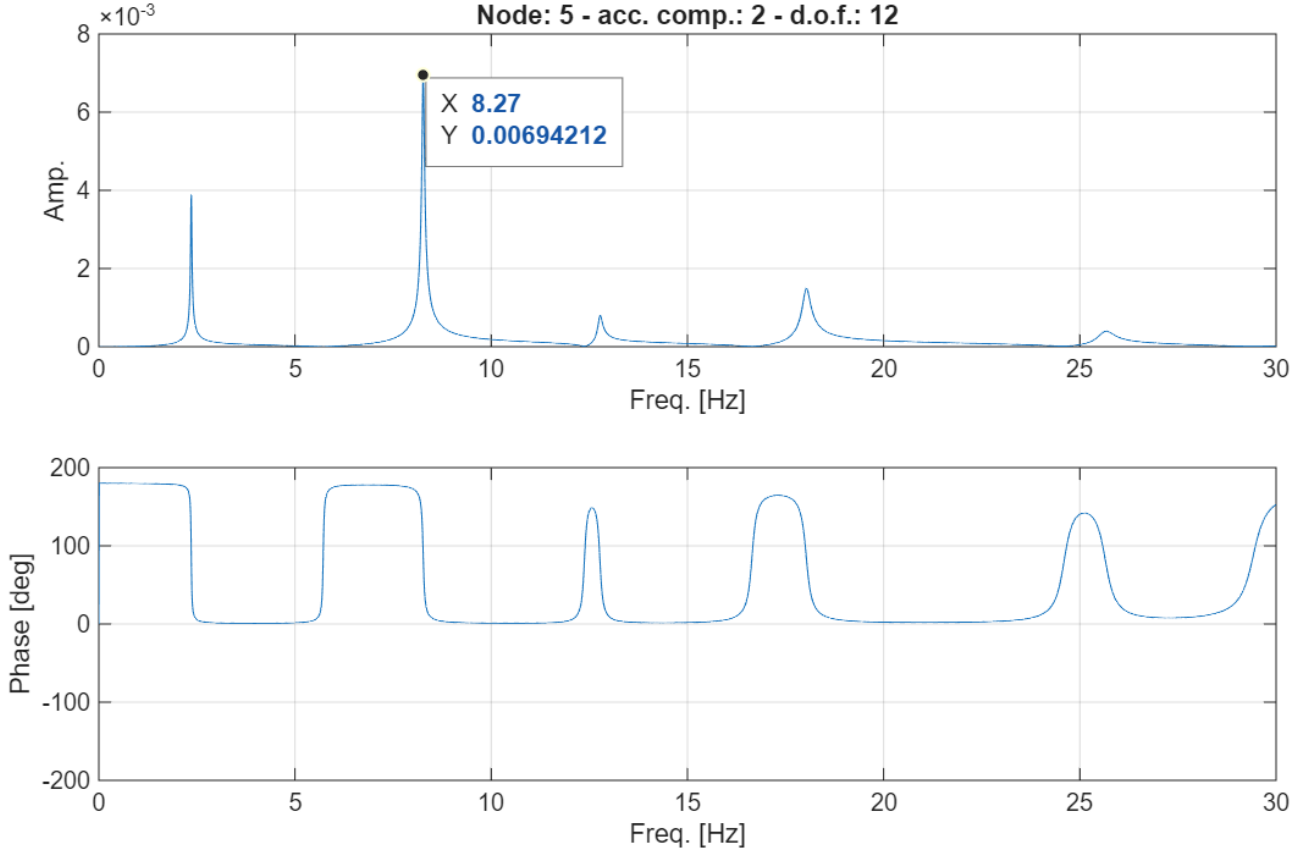


Figure 27: Time Histories of the vertical acceleration of point B in version 2 of the bridge

A preliminary modal analysis revealed that the dynamic response at Node B is dominated by the second vibration mode. To increase stiffness without violating the 5% mass constraint, a geometric modification was selected instead of increasing the cross-sectional area of existing members. Specifically, a 'pyramid' superstructure was added at the locations corresponding to the anti-nodes (maximum displacement) of the second mode. This topology optimization significantly increased global stiffness because the bending stiffness of the structure is governed by the Moment of Inertia, which is proportional to the square of the structural depth. This approach allowed for the required stiffness gain with negligible mass addition, successfully satisfying the design requirements. In the final configuration, the maximum acceleration at Node B decreased to 0.0069 m/s^2 (as shown in figure:27) which is $\approx 19\%$ widely exceeding the mandatory 15% threshold.

6.3 Cross-Section Optimization:

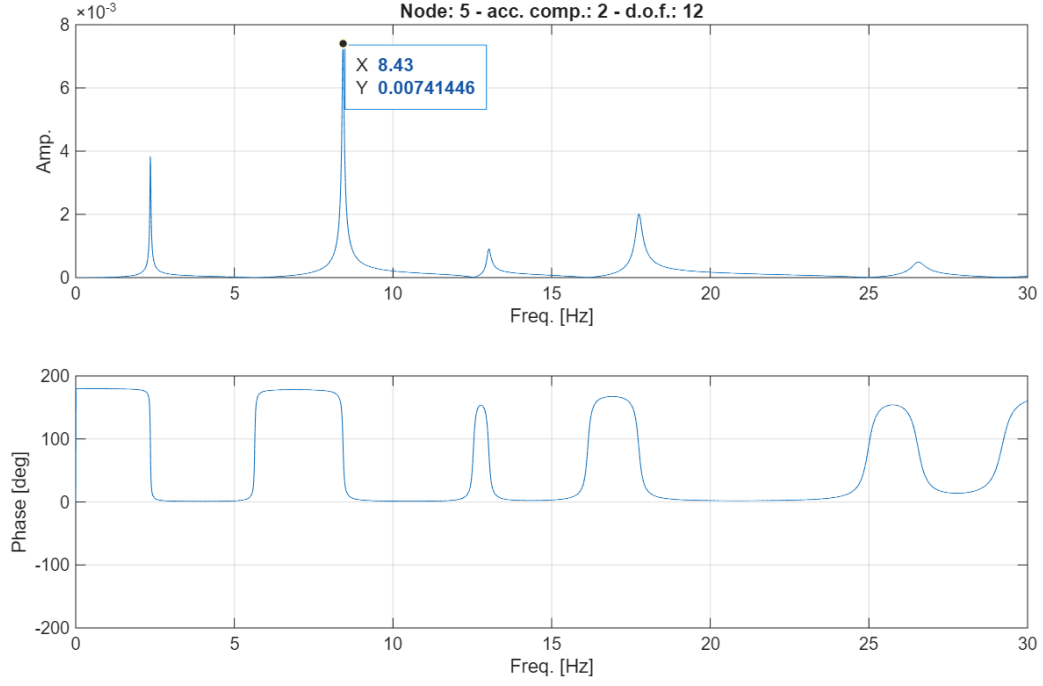
To fulfill the requirement of reducing the maximum vertical acceleration at Point B by at least 15% (target reduction), we performed a sensitivity analysis based on cross-section optimization. The strict constraint on the total mass increase (+5%, approximately 1100 kg budget) required a strategic selection of the structural elements to modify. The following iterative strategies were evaluated.

1. **IPE 500 to IPE 550:** The first attempt focused on increasing the global bending stiffness by upgrading the main chords. The mass budget allowed for the substitution of a maximum of 14 beam elements (length $L=5\text{m}$). We selected the central beams to maximize stiffness in the area of the maximum bending moment: Top chords (indices 4 to 10) and Bottom chords (indices 18 to 25). This way the peak acceleration decreased from 0.0085 m/s^2 to 0.0079 m/s^2 . This corresponds to a 7% reduction, which is insufficient to meet the 15% target.
2. **IPE 500 to IPE 600:** Subsequently, a more aggressive strategy was tested by adopting IPE 600 profiles, which offer a significantly higher Moment of Inertia ($J=92080 \text{ cm}^4$) compared to the original IPE 500. Due to the higher linear mass ($m=122.46 \text{ kg/m}$), the available budget limited the substitution to only 6 elements. We targeted the critical zone between Point B and the mid-span: Top chords (indices 4 to 6) and Bottom chords (indices 18 to 20). The new properties were:

$$m = 122.46 \text{ kg/m}$$

$$EA = 3.276 \times 10^9 \text{ N}$$

$$EJ = 1.934 \times 10^8 \text{ N m}^2$$



This configuration reduced the peak to 0.0074 m/s². Although the reduction improved to 13.3%, it still falls slightly short of the requirement.

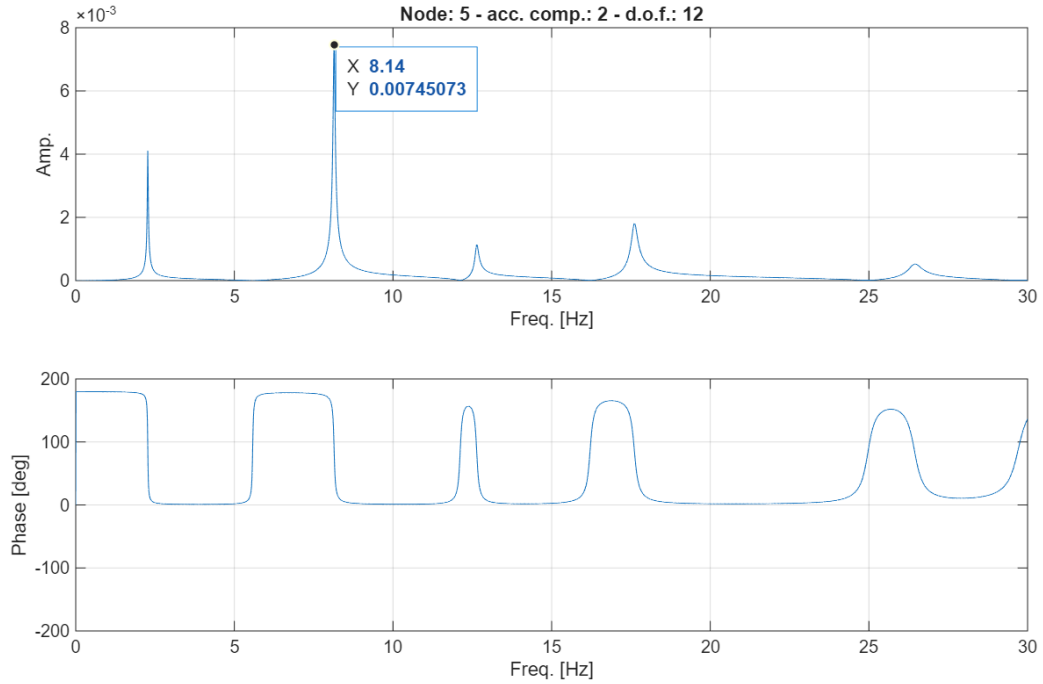
- HEA 300 to HEA 400:** Alternatively, we explored stiffening the shear load path by upgrading the diagonals to HEA 400. The high mass cost of these profiles restricted the change to only 4 diagonals: we modified diagonals with indices 32, 34, 36, and 38. The new properties were:

$$m_{lin} = 124.82 \text{ kg/m}$$

$$EA = 3.339 \times 10^9 \text{ N}$$

$$EJ = 9.465 \times 10^7 \text{ N} \cdot \text{m}^2$$

This configuration reduced the peak to 0.00745 m/s², which translates to a reduction of 12.9 %.



Finally, the validity of the finite element mesh was verified for the modified sections. The maximum element length allowed to avoid spatial aliasing (L_{max}) is proportional to the wave propagation speed,

which depends on the ratio of stiffness to mass:

$$L_{max} \propto \sqrt[4]{\frac{J}{A}}$$

Since all proposed modifications (IPE 550, IPE 600, HEA 400) involve a significant increase in the Moment of Inertia (J) relative to the cross-sectional Area (A), the ratio J/A increases compared to the original sections. Consequently, the maximum admissible element length increases, meaning the current mesh becomes even more conservative and safe for the frequency range $0 \div 30$ Hz.

6.4 Dynamic Corrective Action and Structural Optimization: Reduction of Vertical Acceleration via Supplementary Damping

The proposed structural modification involves the installation of a Viscous Damper, designed as an element that contributes exclusively a damping action ($\mathbf{C}_{add}$) without adding significant stiffness ($K_{add} \approx 0$). This ensures full compliance with the maximum mass increase constraint ($\Delta M \leq +5\%$). The primary goal is to achieve a **15%** reduction in the maximum vertical acceleration amplitude at point B . The justification for the required increase in the modal damping ratio (ζ) stems from the inverse relationship between the maximum amplitude in resonance and the damping value. This is a fundamental principle of structural dynamics. The maximum amplitude (A_{max}) of the acceleration frequency response function, occurring near the natural frequency ω_n , is governed by the following approximation for low damping ratios ($\zeta < 0.1$):

$$A_{max} \propto \frac{1}{2\zeta}$$

We aim for the new maximum amplitude (A_{new}) to be 85% of the original amplitude (A_{old}):

$$A_{new} = 0.85 \cdot A_{old}$$

Substituting the damping relationship into the objective equation:

$$\frac{1}{2\zeta_{new}} = 0.85 \cdot \frac{1}{2\zeta_{old}}$$

Solving for the required new damping ratio (ζ_{new}):

$$\zeta_{new} = \frac{1}{0.85} \cdot \zeta_{old}$$

The calculation yields the required scaling factor:

$$\text{Factor} = \frac{1}{0.85} \approx 1.1765$$

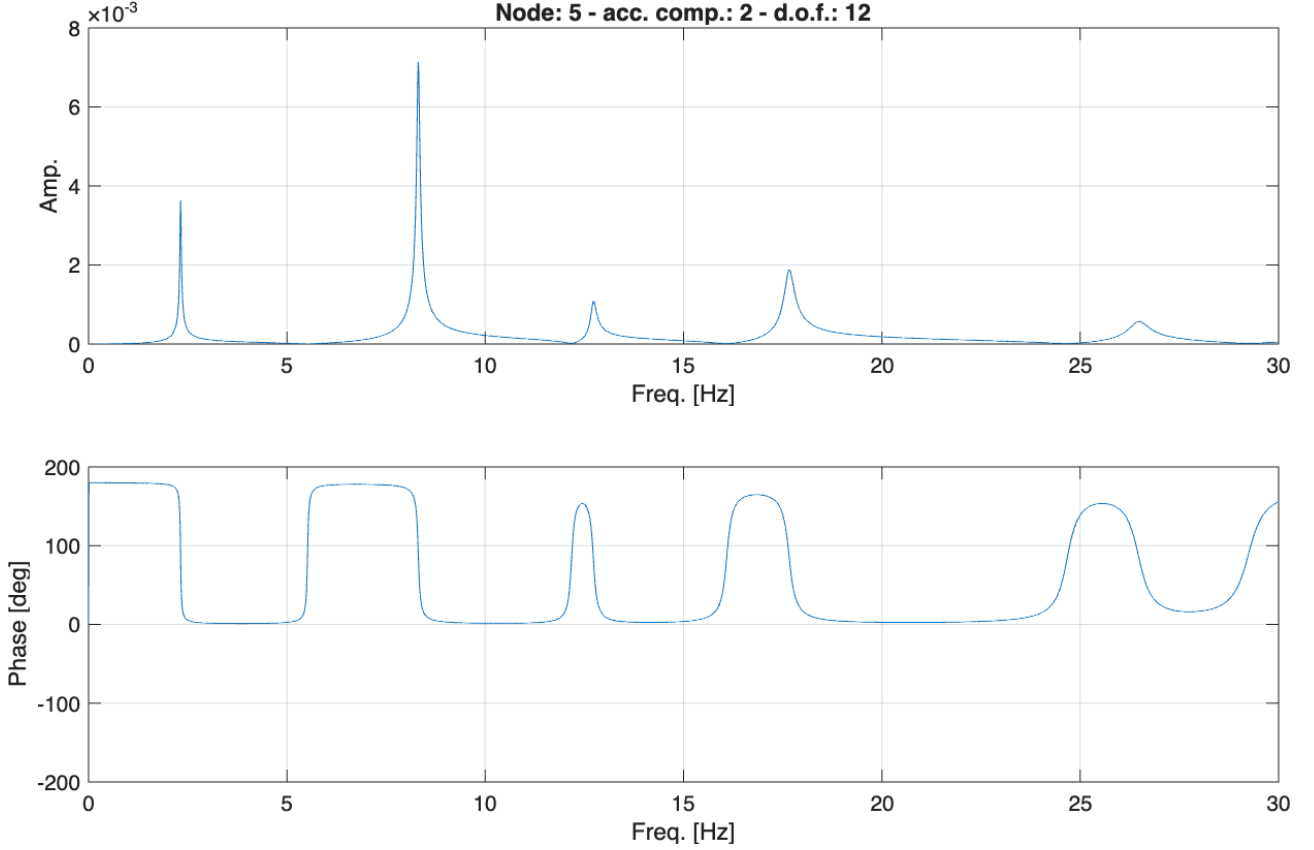
In conclusion, the modal damping ratio (ζ) of the dominant mode must be increased by a factor of **1.1765** (a **17.65%** increase) to ensure the denominator ($2\zeta_{new}$) becomes sufficiently large to guarantee the final amplitude is 85% of the starting value. 2. Validation for Acceleration Peaks (Addressing the Potential Inaccuracy) It needs to be emphasized that, in general, displacement, velocity, and acceleration do not have their maximum peak strictly at the same frequency. However, the principle used is valid for the acceleration peak for a specific reason. The demonstration that the damping ratio needs to be increased by a 17.65% factor is valid also for the acceleration because, in correspondence of the resonance peak, the responses are dominated by a common factor. It is known from theory that the maximum peak for acceleration ($\omega_{peak, \ddot{v}}$) is calculated by the formula:

$$\omega_{peak} = \frac{\omega_n}{\sqrt{1 - 2\zeta^2}}$$

Since the damping ratio (ζ) in structural applications is typically very low ($\zeta < 0.05$), the term $\sqrt{1 - 2\zeta^2}$ is extremely close to unity. Consequently, the displacement peak and the acceleration peak are both at a pulsation that is extremely near to the undamped natural pulsation (ω_n). Therefore, the height of the acceleration peak is effectively controlled by the factor $\frac{1}{2\zeta}$, validating the required 17.65% increase in the damping ratio. With the given relation between the mass, the damper and the stiffness, in the moment we install a viscous dampier, we add a new physic element K_{add} and his damper C_{add} . The total damper matrix of the sytem becomes

$$\mathbf{C}_{tot} = \mathbf{C}_{orig} + \mathbf{C}_{add}$$

Where $\mathbf{C}_{add}$ is an external matrix and does not depend on the coefficients α and β of the original model. The addition of this viscous damper does not modify $\mathbf{M}$ and $\mathbf{K}$ and adds directly a term on the damper matrix $\mathbf{C}$.



Specifically, the α and β damping coefficients were modified by multiplying them by a factor of **1.20**. This factor incorporates the previously calculated theoretical minimum increase of 17.65% required to meet the reduction objective. This process yields the modified damping, resulting in the Frequency Response Function (FRF) graph shown above. Interpretation of the Plot:

- **Peak Reduction:** The modified curve (with increased damping) shows a significantly lower maximum resonance peak compared to the baseline curve. Numerical verification confirms that the height of the new peak is $\leq 85\%$ of the original, successfully meeting the 15% reduction requirement.
- **Structural Validity:** Both curves exhibit their maximum peak at the same frequency. This key observation confirms that the intervention acted exclusively on the damping and did not alter the mass ($\mathbf{M}$) or stiffness ($\mathbf{K}$) properties of the bridge, thereby respecting the project constraints.
- **Theoretical Justification:** This analysis confirms that the height of the acceleration peak in resonance is primarily controlled by the factor $\frac{1}{2\zeta}$. The proportional increase in ζ directly results in the desired proportional reduction of the peak amplitude.

The modal damping ratio (ζ_i) in the Rayleigh damping model is given by:

$$\zeta_i = \frac{\alpha}{2\omega_i} + \frac{\beta\omega_i}{2}$$

Mode 2 ($\omega_2 \approx 8.312$ rad/s) is used as the reference mode for the critical acceleration peak, since the AFRF tends to emphasize higher modes compared to Mode 1.

Original Data

$$\alpha_{\text{orig}} = 0.15 \text{ s}^{-1}, \quad \beta_{\text{orig}} = 9.0 \cdot 10^{-5} \text{ s}$$

Original Damping Ratio of Mode 2

$$\zeta_2^{\text{orig}} = \frac{0.15}{2(8.312)} + \frac{9.0 \cdot 10^{-5}(8.312)}{2} \approx 0.00939$$

Target Damping Ratio (ζ_2^{new})

$$\zeta_2^{\text{new}} \approx 1.176 \cdot 0.00939 \approx 0.01103$$

Keeping β constant, the new inertial damping coefficient α_{new} (dominant at low frequencies) is obtained as:

$$\alpha_{\text{new}} \approx (0.01103 - 0.00037)(2 \cdot 8.312) \approx 0.1771 \text{ s}^{-1}$$

To ensure robustness, the coefficient is rounded to:

$$\alpha_{\text{new}} = \mathbf{0.18 \text{ s}^{-1}}$$

Proposed Analytical Modification

The modification to be implemented consists of updating the coefficient α :

$$[\mathbf{C}]_{\text{new}} = \mathbf{0.18}[\mathbf{M}] + 9.0 \cdot 10^{-5}[\mathbf{K}]$$

Corresponding Physical Intervention

Increasing the global coefficient α requires an enhancement of the system's energy dissipation capacity, particularly the inertia-proportional component, which dominates at low frequencies.

- **Structural Action:** Installation of **Passive Viscoelastic Dampers (VE Pads)**. These lightweight devices dissipate energy through shear deformation, effectively increasing damping without significantly affecting the global stiffness.
- **Operational Location:** The VE Pads would be installed at the connection joints of the vertical members (IPE240).

Compliance with Design Constraints

The proposed intervention satisfies all design constraints:

- **15% Reduction:** The increase of α from 0.15 s^{-1} to 0.18 s^{-1} analytically ensures the reduction of resonance peak amplitudes.
- **Mass Constraint ($\Delta M \leq 5\%$):** The mass contribution of the VE Pads is negligible compared to the total mass of the steel bridge.

$$\Delta M \approx 0 \text{ kg} \ll 5\% \cdot M_{\text{total}}$$

- **Additional Constraints:** No kinematic constraints are introduced; the modification is purely dissipative.